A fixed-target approach to the sharp iid Berry–Esseen inequality*

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Abstract

We study the conjectured sharp iid Berry–Esseen constant $C_{\mathrm{E}} = (3+\sqrt{10})/(6\sqrt{2\pi})$ by maximizing excess over this fixed target. Attainment, contact and a conservation identity reduce a possible counterexample to one actual marginal with a common moment-defect budget. An explicitly stated Fourier certificate supplies the coarse entry. For $n \geq 700$, elementary moment inequalities and two contact rays then acquire one probability and support strip. A single-count fourth-moment argument handles small conditional noise; five counts and two concave quartics handle the remaining noise. Both retain the actual marginal, all threshold phases and atoms, and the complete incoming tails. The resulting acquired-witness exclusion uses the true sharp predecessor. The global implication is conditional on a genuine finite base through 699 and the complete entry predicates. Neither that base nor the full conjecture is proved here. This manuscript integrates the one-strip reductions and their analytical proofs in a single route.

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*This paper was written with substantial assistance from AI systems; see the end of Section 8.

1 The problem and the selected route

Let $\mathcal{P}$ be the laws on $\mathbb{R}$ with mean zero, variance one and finite third absolute moment $\rho(P) = \mathbb{E}|X|^3$. For independent copies $X_1, \dots, X_n$ of P, put $S_n = \sum_i X_i$ and

$$\Delta_n(P) = \sup_{H \in \mathbb{R}} |\mathbb{P}_P(S_n \le H) - \Phi(H/\sqrt{n})|, \qquad C_n = \sup_{P \in \mathcal{P}} \frac{\sqrt{n}\,\Delta_n(P)}{\rho(P)}.$$

Here Φ and ϕ are the standard normal distribution function and density. The sharp-constant conjecture is

$$\sup_{n \ge 1} C_n = C_{\mathrm{E}}, \qquad C_{\mathrm{E}} = \frac{3 + \sqrt{10}}{6\sqrt{2\pi}} = 0.409732\ldots. \tag{1}$$

Throughout, H is a raw sum threshold; terminating decimals in inequalities are exact rationals. Affine standardization gives the equivalent result for any mean and positive variance, with third absolute central moment divided by the cube of the standard deviation.

Berry and Esseen established the classical order $n^{-1/2}$ [3, 5, 6]. Esseen's lower bound [7] explains the proposed constant. If $p_* = (4 - \sqrt{10})/2$, $q_* = 1 - p_*$ and $X = (B - p_*)/\sqrt{p_* q_*}$ for $B \sim \mathrm{Bernoulli}(p_*)$, the fixed-law lattice limit gives

$$\lim_{n \to \infty} \frac{\sqrt{n}\Delta_n(P)}{\rho(P)} = \frac{3 + q_* - p_*}{6\sqrt{2\pi}(p_*^2 + q_*^2)} = C_{\mathrm{E}}. \tag{2}$$

Indeed $p_*^2 + q_*^2 = 10 - 3\sqrt{10}$ and $(3 + q_* - p_*)/(p_*^2 + q_*^2) = 3 + \sqrt{10}$. Every constant below C_{E} therefore fails for a sufficiently large sum of this one law. The upper conjecture is associated with Zolotarev [22, 23]; Schulz proves the strict finite-index bound for every nondegenerate Bernoulli law [13].

Context.

Fourier and structural moment inequalities progressively improved the universal upper bound [11, 14, 16, 18, 19]. We use Shevtsova's structural inequality explicitly at entry. Recent manuscripts report smaller upper constants [10, 20], and He and Cheng report eventual sharpness beyond a uniform, very large threshold [9]. Work on independent arrays and extremizers addresses the related non-iid problem [1, 2]. Tao's registry and the available formalization projects distinguish numerical certificates, analytical inputs and formalized implications [4, 17, 21]. None of these recent reported bounds is a premise below. An earlier AI-assisted working draft is on Hexagon [12].

Fix the target rather than an unknown ratio.

At a fixed n, maximize

$$a(P, H) = \sqrt{n}\{\mathbb{P}_P(S_n \le H) - \Phi(H/\sqrt{n})\} - C_{\mathrm{E}}\rho(P). \tag{3}$$

The moment penalty yields attainment and compact-support contact without assuming uniformly integrable third moments. Integrating contact against the zero-bias density gives a common budget for excess, support-gap defect and leakage. The transform originates in [8]. A complete coarse Fourier entry supplies $nD \le 32.20665378$ for the same law, under its true sharp predecessor.

For $n \ge 700$, direct inequalities give $3/8 < p < 93/200$, inward displacement below $1/250$ and conditional noise below $7/20$. Two contact rays acquire one support strip. A single-count estimate for noise at most $1/100$ and a five-count estimate above it then give strict negative excess. The shared endpoint and zero noise are included.

The finite boundary.

This is a conditional acquired-witness exclusion, not an unconditional cutoff at 700. A least failure supplies the required predecessor only if all earlier indices have been controlled. A genuine base through 699, together with the complete entry predicates, implies the sharp conjecture; Section 7 proves this implication. A base only through 41 locates a first failure in $42, \dots, 699$ but does not rule it out. Those finite bounds remain open in this work.

1.1 Reading the argument

The proof follows one chain: fixed-target maximum, contact and budget, coarse entry, one support strip, and count comparison. The first step selects one actual marginal; no later relaxation changes that marginal. Reflection selects positive closed-event orientation before entry, so its signed defect is acquired in the orientation where it is used. The complete Fourier predicates enter only at coarse entry. The later estimates are analytical, with finite rational inequalities specified in the proof.

For $B = \mathbf{1}_{\{X>0\}}$, put $p = \mathbb{P}(B = 1)$, $q = 1 - p$, $v = pq$ and $m = \mathbb{E}X_+ = \mathbb{E}X_- > 0$. If $W = X - \mathbb{E}(X \mid B)$ and $\ell = m/v$, then

$$X = \ell(B + U - p), \quad U = W/\ell, \quad \theta = \mathbb{E}W^2, \quad s = \mathbb{E}U^2, \quad \lambda = ns = \frac{n\theta}{\ell^2}.$$

Thus θ and λ use different units; $v + s = \ell^{-2}$. The inward displacements a_0, a_1 are also in count coordinates, and $d = a_0 + a_1$. The physical defect is the distinct symbol $D = (3 + \sqrt{10})\rho - \gamma - 3J$, where $\gamma = \mathbb{E}X^3$, $\beta = \mathbb{E}[X|X|]$ and J is the weighted support-gap sum. Reconstructing X from $B + U$ leaves the raw threshold H unchanged.

The main text gives the variational, geometric, entry and comparison arguments. The appendices contain the Fourier enclosure used at entry and elementary Gaussian constants. Alternative smaller-index and local routes remain in the repository, outside this selected compilation.

2 Fixed-target extremizers and the conserved resource

We optimize the excess above a fixed candidate constant. This gives a variational problem whose coefficient is known before any acquisition step. The positive excess is retained through the contact calculation and the conserved identity; it becomes a nonnegative term in the final resource. No upper bound for the maximizing normalized score is needed. Throughout, sums have independent identically distributed summands with mean zero, variance one and finite third absolute moment ρ. Fix $n \geq 2$, $c > 0$, and a real raw sum threshold H. Set

$$z = H/\sqrt{n}, \quad F = \mathbb{P}(S_n \leq H), \quad a = \sqrt{n}\{F - \Phi(z)\} - c\rho,$$
$$\varphi = \phi(z), \quad \beta = \mathbb{E}[X|X|], \quad \gamma = \mathbb{E}X^3.$$

In particular a is not divided by ρ.

2.1 Attainment, contamination and the entire closed support

Proposition 2.1. *For each fixed n and $c > 0$, the supremum of a over these laws and thresholds is attained, and the closed-CDF discrepancy at every attaining pair is positive. At every maximizing pair,*

$$I(y) = n^{3/2}\{h(y) - F\} + ny\varphi + \tfrac{1}{2}H\varphi(y^2 - 1) + \tfrac{3}{2}c\rho(y^2 - 1) - c\{|y|^3 - \rho - 3y\beta\} \leq 0, \qquad h(y) = \mathbb{P}(S_{n-1} \leq H - y), \tag{4}$$

with equality on the whole closed support. That support is compact. Neither statement assumes $a \geq 0$ or $H \geq 0$.

Proof. Write A for the supremum. Since $\rho \geq 1$ and the discrepancy is at most one, $A \leq \sqrt{n}-c$. Symmetric Bernoulli evaluated at $H = n$ gives $A \geq \sqrt{n}(1 - \Phi(\sqrt{n})) - c > -c$. For a maximizing sequence, $a_j \to A$ and $c\rho_j \leq \sqrt{n} - a_j$ give a common third-moment ceiling. The discrepancies satisfy $\sqrt{n}(F_j - \Phi(H_j/\sqrt{n})) = a_j + c\rho_j \geq a_j + c > \eta$ eventually for some $\eta > 0$. Chebyshev bounds the negative threshold tail by n/H_j^2; the positive discrepancy is at most the normal tail, also at most n/H_j^2. Thus H_j is bounded.

Tightness gives a subsequence $P_j \Rightarrow P$, $H_j \to H$ and $\rho_j \to b$. The common third-moment ceiling M gives tail bounds $\int_{|x|>K} x^2 dP_j \leq M/K$ and $\int_{|x|>K} |x| dP_j \leq M/K^2$. Bounded continuous truncations therefore preserve mean and variance. Lower semicontinuity gives $1 \leq \rho(P) \leq b$. For fixed n, the sum laws converge weakly. Applying the closed-set inequality first at $H + t$ and then letting $t \downarrow 0$ gives $\limsup F_j(H_j) \leq F_P(H)$, including an atom at H. Consequently

$$a(P, H) \geq A + c\{b - \rho(P)\} \geq A.$$

Maximality forces equality and $\rho(P) = b$. Convergence of third moments is a conclusion here; it was not inferred from bounded third moments. The discrepancy remains positive since $A + c\rho \geq A + c > 0$.

For contact, contaminate the raw marginal by $(1 - \epsilon)P + \epsilon\delta_y$. Its mean is ϵy, variance $v_\epsilon = 1 + \epsilon(y^2 - 1) - \epsilon^2 y^2$, and centered third absolute moment has derivative $|y|^3 - \rho - 3y\beta$ at zero. The last assertion follows by dominated differentiation of $|x - \epsilon y|^3$, with bound $3|y|(|x| + |y|)^2$. Evaluate the standardized law at the standardized image of the fixed raw event $S_n \leq H$. The raw event probability is a polynomial in ϵ, with derivative $n(h(y) - F)$ even at an atom collision. The normal argument is $(H - n\epsilon y)/\sqrt{nv_\epsilon}$, whose derivative is $-\sqrt{n}y - H(y^2 - 1)/(2\sqrt{n})$. The one-sided derivative of the excess is exactly (4); maximality makes it nonpositive.

Its integral against P is zero. All terms are integrable. The closed-event map h is upper semicontinuous; at an atom it takes the larger one-sided value. Thus I is upper semicontinuous. Strict negativity at a closed-support point would hold with a uniform negative margin on a neighborhood of positive P-mass, contradicting its zero integral. Finally $I(y) \leq -c|y|^3 + C_2 y^2 + C_1|y| + C_0$ for constants depending on this pair. It is strictly negative on both sufficiently distant rays, proving compact support. No higher moments were used before this step. □

2.2 Reflection, equality, and the first unresolved index

The sign of the attained maximum detects failure of a strict row bound, including equality. If C_n denotes the supremum of the normalized absolute discrepancy, reflection gives, including an atom at H,

$$\mathbb{P}(-S_n \leq -H) - \Phi(-z) = \Phi(z) - \mathbb{P}(S_n < H) \geq \Phi(z) - F.$$

Thus each negative closed-CDF error is bounded by a positive closed-CDF error of the reflected law; equality at an atom is not assumed. Consequently C_n is also the supremum of positive closed scores r. Since $a = \rho(r - c)$, $C_n < c$ gives $A \leq C_n - c < 0$, and $C_n > c$ gives $A > 0$. If $C_n = c$, choose $r_j \uparrow c$ with $r_j \geq c/2$. Then $\rho_j \leq 2\sqrt{n}/c$, so $\rho_j(r_j - c) \to 0$ and $A = 0$. Hence $C_n \geq c$ supplies an attained pair with $a \geq 0$.

Corollary 2.2 (First failure of a strict bound)**.** *Assume a genuine finite base $C_j \leq c$ for $1 \leq j < N$. If a strict bound $C_n < c$ fails at any index $n \geq N$, there is a least such index, an attained fixed-target pair with $a \geq 0$, and true predecessor bounds $C_j \leq c$ for every $j < n$. The bounds are strict for $N \leq j < n$.*

Proof. The nonempty set of failing indices at least N has a least member. The preceding equivalence supplies its maximizing pair. Indices below N use the finite base; the other earlier indices use minimality. □

Thus an argument excluding every such pair for $n \geq N$ would give strict bounds above the finite base, including exclusion of equality. The subsequent entry argument uses only $C_{n-1} \leq c$; it does not use an upper bound for the actual score $c + a/\rho$. The finite base and the eventual exclusion remain separate obligations.

Equivalently, $\Delta = Q - h \geq 0$ everywhere and $\Delta = 0$ on the closed support, where

$$Q(y) = F + \frac{H\varphi + c\rho}{2n^{3/2}} - \frac{n\varphi + 3c\beta}{n^{3/2}}y - \frac{H\varphi + 3c\rho}{2n^{3/2}}y^2 + \frac{c}{n^{3/2}}|y|^3. \tag{5}$$

2.3 Zero-bias integration and the exact identity

We now allow an arbitrary compactly supported standardized law, retaining the definitions of a, Q, h, Δ above without assuming contact. Put $\nu = \mathcal{L}(H - S_{n-1})$ and $\tau(t) = \mathbb{E}[X\mathbf{1}_{X \geq t}]$. Then $\tau \geq 0$, $\int \tau = 1$. Over the bounded complementary intervals (u, v) of the closed support K, define

$$J = \sum \tau_{u,v}(v-u)^2, \qquad J_1 = \sum \tau_{u,v} \int_u^v (t-u)t\, dt,$$
$$J_2 = \sum \tau_{u,v} \int_u^v (t-u)t|t|\, dt, \qquad L = \sum \tau_{u,v} \int_{[u,v)} (t-u)\, d\nu(t).$$

Here $\tau_{u,v}$ is the constant interior value. Introduce two contact residuals:

$$\xi = \mathbb{E}[X\Delta(X)], \qquad B = \int_K \tau(t)\Delta(t)\, dt + \sum_{(u,v)} \tau_{u,v}(v-u)\Delta(v). \tag{6}$$

The integral over K is a Lebesgue integral, not an expectation under P. It must be retained when the closed support has positive Lebesgue measure. If $K \subset [-R, R]$, then $J \leq 2R$, $|J_1| \leq RJ/2$, $|J_2| \leq R^2 J/2$, and $0 \leq L \leq 2R$. Also Δ is bounded on the convex hull, so the sum in B is absolutely convergent because $\sum \tau_{u,v}(v-u) \leq 1$.

Let $M_4 = \mathbb{E}X^4$, $M_5 = \mathbb{E}|X|^5$, and put

$$K_q = M_5/4 - \rho M_4/2 - 3\beta\gamma/2 + \rho/2, \qquad K_1 = \mathbb{E}[X|X|^3] - 3\beta - 3\rho\gamma/2,$$

$$\Omega = J_1 + (M_4 - 3)/6, \quad \varepsilon = \mathbb{E}q_z(S_n/\sqrt{n}) - \mathbb{E}q_z(Z_0) - 2\gamma\varphi/(3\sqrt{n}), \quad q_z(x) = (x-z)|x-z|,$$

where Z_0 is standard normal. Define $E_n = -n^{3/2}\varepsilon/2$ and

$$\mathcal{R}_c = c(3\beta J/2 + 3\rho J_1 - 3J_2 - K_q) + H(cK_1 + \varphi\Omega) - H^2(c\rho + \gamma\varphi/2) + E_n. \tag{7}$$

Theorem 2.3 (Conservation with contact residuals)**.** *For every such law, every $H \in \mathbb{R}$ and every $c > 0$,*

$$(n + H^2)a + nc\rho - \frac{n\varphi}{6}(\gamma + 3J) + n^{3/2}L = \mathcal{R}_c + n^{3/2}(B - H\xi). \tag{8}$$

At a fixed-target maximizing pair, $B = \xi = 0$. In particular,

$$(n + H^2)a + nc\rho - \frac{n\varphi}{6}(\gamma + 3J) + n^{3/2}L = \mathcal{R}_c. \tag{9}$$

The coefficient of the unnormalized excess is $n + H^2$.

Proof. There are two direct evaluations of $\mathcal{A} = \int \tau\Delta$. Since $h(t) = \nu([t,\infty))$, one has $h(t) = h(v) + \nu([t,v))$ inside a gap. Thus bounded Fubini gives

$$\int_u^v \Delta(t)dt = (v-u)\Delta(v) + \int_u^v (t-u)(-Q'(t))dt - \int_{[u,v)} (t-u)d\nu(t).$$

An atom at v belongs to $h(v)$ and is not in the last integral; an atom at u is included but multiplied by zero. Summing, including the support integral in (6), gives

$$\mathcal{A} = \frac{\varphi J}{2\sqrt{n}} + \frac{3c\beta J/2 + (H\varphi + 3c\rho)J_1 - 3cJ_2}{n^{3/2}} - L + B. \tag{10}$$

For the second evaluation, signwise Fubini proves $\int \tau g' = \mathbb{E}[Xg(X)]$ for absolutely continuous g with bounded derivative on the relevant compact range. In particular

$$\int t\tau = \gamma/2, \quad \int t^2\tau = M_4/3, \quad \int |t|^3\tau = M_5/4.$$

Apply that same identity conditionally to $g_s(x) = \min(s+x, H)$ and then use exchangeability. The choice of derivative at its single corner is immaterial under the density τ. It follows directly that

$$\int \tau(t)h(t)dt = \mathbb{E}[X_n \min(S_n, H)] = \frac{1}{n}\mathbb{E}[S_n \min(S_n, H)] = T,$$

where $T = \mathbb{E}[(Z^2 - zZ)\mathbf{1}_{Z\le z}]$, $Z = S_n/\sqrt{n}$. The last expression vanishes at an endpoint atom. Integrating (5) against τ therefore yields

$$\mathcal{A} = F - \frac{\gamma\varphi}{2\sqrt{n}} + \frac{H\varphi(3-M_4)/6 + cK_q}{n^{3/2}} - T. \tag{11}$$

Set $M = \mathbb{E}[Z\mathbf{1}_{Z\le z}]$. Since $\mathbb{E}[Xh(X)] = M/\sqrt{n}$, integration of $XQ(X)$ against P gives

$$M + \varphi = -\frac{\gamma H\varphi}{2n} + \frac{cK_1}{n} - \sqrt{n}\xi. \tag{12}$$

The pointwise identity $q_z(x) = (x-z)^2 - 2(x-z)^2\mathbf{1}_{x\le z}$, with the same identity for a normal variable, gives

$$T - \Phi(z) = -\varepsilon/2 - \gamma\varphi/(3\sqrt{n}) + z(M+\varphi) - z^2(F - \Phi(z)).$$

Inserting (12) and $\sqrt{n}(F-\Phi(z)) = c\rho + a$ into (11) gives

$$\begin{aligned}\sqrt{n}\mathcal{A} &= c\rho + a - \gamma\varphi/6 + P_0/n + \sqrt{n}\varepsilon/2 + H\sqrt{n}\xi,\\ P_0 &= H^2(c\rho + a + \gamma\varphi/2) - H\{cK_1 + \varphi(M_4-3)/6\} + cK_q.\end{aligned}$$

Equate this with (10) multiplied by $\sqrt{n}$, then multiply by n and collect terms. This is (8). Under closed-support contact, every term defining B and ξ vanishes. □

The compensated Fourier representation of this same E_n is proved in Appendix A. Its comparator has transform $e^{-nt^2/2}(1 - in\gamma t^3/6)$. The residual identity also explains why applying the residual-free formula to arbitrary laws would be invalid. For example, let X be uniform on $[-\sqrt{3}, \sqrt{3}]$ and take $H = 0$. There are no support gaps, $F = T = 1/2$, $\beta = \gamma = 0$, and $K_q = 3\sqrt{3}/40$. Therefore $B = \int_K \tau\Delta = cK_q/n^{3/2} > 0$. Omitting the support integral would give a false identity, even though the gap sums are all empty. This is an off-contact negative control, not a counterexample to the contact theorem.

2.4 A positive resource and its ordered phase price

Put $m = \mathbb{E}X_+ = \mathbb{E}X_- > 0$ and $\sigma = \sqrt{10}$. Signwise Cauchy gives $a_\pm^2 \le m b_\pm$ and $m^2 \le a_+ a_-$, where $a_\pm = \mathbb{E}X_\pm^2$ and $b_\pm = \mathbb{E}X_\pm^3$. Therefore $b_+ b_- \ge m^2$, or $\gamma^2 + 4m^2 \le \rho^2$. A gap on one side of zero has $(v-u)^2 \le 2\int_u^v |t|dt$. If a central gap $(-A, B)$ exists, its extra contribution is $2mAB$, and $0 \le \mathbb{E}[(X+A)(X-B)] = 1 - AB$. Signwise Tonelli then gives $J \le \rho + 2m$. It follows that

$$D = (3+\sigma)\rho - \gamma - 3J = 3(\rho + 2m - J) + (\sigma\rho - \gamma - 6m) \ge 0. \tag{13}$$

The last term is nonnegative by Cauchy in two dimensions. These statements hold with or without an atom at zero.

Specialize to $c = C_{\mathrm{E}} = (3+\sigma)\phi(0)/6$ and contact. With $w = H^2/(2n)$, $e = e^{-w}$, the identity becomes

$$\mathcal{R}_c = (n + H^2)a + nc\rho(1-e) + \frac{n\phi(0)e}{6}D + n^{3/2}L. \tag{14}$$

If $a \ge 0$, every term is nonnegative. This specialization to C_{E} is essential for this definition of D.

Corollary 2.4 (Raw bound, density, and phase price)**.** *At such a contact pair with $a \ge 0$, assume $n \ge N \ge 2$, $\rho \ge l > 0$, $0 < c_- \le c$, and a proved raw bound $\mathcal{R}_c \le T < Nc_-l$. A negative T excludes the pair. Otherwise*

$$\gamma + 3J > 0, \qquad e \ge 1 - T/(Nc_-l) > 0,$$
$$\mathcal{U} := (n+H^2)a + n\phi(0)D/6 + n^{3/2}L \le \mathcal{R}_c,$$
$$\mathcal{U} \le \mathcal{R}_c - \tfrac{1}{2}(c_-l - T/N)H^2.$$

No condition $D < 2$ or own-score upper bound is required.

Proof. Set $M_c = c\rho - \phi(0)D/6 = \phi(0)(\gamma + 3J)/6$. The general master identity gives

$$eM_c = (1 + H^2/n)a + c\rho + \sqrt{n}L - \mathcal{R}_c/n \ge c\rho - T/n > 0.$$

Thus $M_c > 0$ and, since $D \ge 0$, $M_c \le c\rho$. The density bound follows. Regrouping (14) gives $\mathcal{R}_c = \mathcal{U} + n(1-e)M_c$, proving the unweighted bound. Finally

$$n(1-e)M_c = n(e^w - 1)eM_c \ge \frac{H^2}{2}(c\rho - T/n) \ge \frac{H^2}{2}(c_-l - T/N).$$

This prices one term once. It is a bound on $\mathcal{U}$, not an improved bound on the raw remainder. □

Every conclusion holds at each actual index and raw threshold. Common bounds on a closed moment cell make the result uniform even when the maximizing law varies with n. There is no exchange of a limit with contact, support gaps, moments, or an integral. In particular, the order of use is essential: a raw ceiling first proves $M_c > 0$ and a density floor; only then may the Gaussian phase term be subtracted to improve $\mathcal{U}$. A smaller resource ceiling is not a replacement for the raw ceiling in that licensing step.

2.5 Two elementary moment domains

The moment inequalities below are used before any finite-dimensional relaxation. They hold for the actual marginal, without a bounded-support assumption. In particular, they are unaffected by adding a very small atom far from the origin.

Theorem 2.5 (Signed and absolute third moments)**.** *If* $\mathbb{E}X = 0$, $\mathbb{E}X^2 = 1$, $\rho = \mathbb{E}|X|^3 < \infty$, *and* $\gamma = \mathbb{E}X^3$, *then*

$$\rho \geq \frac{\gamma^2 + 2}{\sqrt{\gamma^2 + 4}}.$$

This is the sharp moment relation of Shevtsova [15]; in the cited arXiv version its equivalent form is Theorem 2. We include a factorization proof to specify the domain used by the finite certificates.

Proof. Set $s = \sqrt{\gamma^2 + 4}$, $a = (s - \gamma)/2$, $b = (s + \gamma)/2$ and $c = \gamma(\gamma^2 + 6)/s^3$. Then $a, b > 0$, $ab = 1$ and $a + b = s$. Define

$$w(y) = |y|^3 - cy^3 - \frac{12y^2}{s^3} + \frac{6\gamma y}{s^3} + \frac{4}{s^3}.$$

For $y, t \geq 0$, direct multiplication gives

$$w(y) = \frac{2a^2}{s^3}(y - b)^2\{2 + (a + 3b)y\}, \qquad w(-t) = \frac{2b^2}{s^3}(t - a)^2\{2 + (3a + b)t\}.$$

Thus $w \geq 0$ on the whole real line. Taking expectations and using the first three signed moments gives

$$0 \leq \mathbb{E}w(X) = \rho - c\gamma - 8/s^3 = \rho - (\gamma^2 + 2)/s.$$

All terms are integrable under the stated third-moment hypothesis. □

Lemma 2.6 (The signed second moment)**.** *With the same normalization and* $\beta = \mathbb{E}[X|X|]$,

$$|\beta| < 1, \qquad \rho \geq \frac{1 + \beta^2}{\sqrt{1 - \beta^2}}.$$

Proof. Both signs have positive probability. Put $m = \mathbb{E}X_+ = \mathbb{E}X_- > 0$ and $a_\pm = \mathbb{E}X_\pm^2 = (1 \pm \beta)/2 > 0$. If $p_\pm = \mathbb{P}(\pm X > 0)$, Cauchy's inequality gives $m^2 \leq p_\pm a_\pm$. Since $p_+ + p_- \leq 1$, this implies

$$m^2(1/a_+ + 1/a_-) \leq 1, \qquad 2m \leq \sqrt{1 - \beta^2}.$$

Another application of Cauchy's inequality, to $X_\pm^{1/2}$ and $X_\pm^{3/2}$, gives $m\mathbb{E}X_\pm^3 \geq a_\pm^2$. Summing yields $m\rho \geq (1 + \beta^2)/2$, which proves the claim. A possible zero atom enters only through $p_+ + p_- \leq 1$ and causes no exception. □

3 The two-point skeleton and its moment defect

Every centered law can be separated into its positive and nonpositive parts. Replacing each part by its conditional mean gives a two-point law; the remaining conditional fluctuations measure its departure from that law. We first make this decomposition exact, then identify a moment defect that vanishes at Esseen's extremal distribution.

3.1 The sign split and its units

Fix $P \in \mathcal{P}$ and put

$$p = \mathbb{P}(X > 0), \quad q = 1 - p, \quad v = pq, \quad m = \mathbb{E}X_+ = \mathbb{E}X_- > 0, \quad B_0 = \mathbf{1}_{\{X>0\}}.$$

Both p, q are positive; a possible atom at zero is assigned to the nonpositive class. Define $V = \mathbb{E}[X \mid B_0]$ and $W = X - V$, so that

$$V = -m/q \quad (B_0 = 0), \qquad V = m/p \quad (B_0 = 1), \qquad \theta = \mathbb{E}W^2 = 1 - m^2/v < 1.$$

Write $\theta_- = \mathbb{E}[W^2; B_0 = 0]$ and $\theta_+ = \mathbb{E}[W^2; B_0 = 1]$. The distance between the two conditional means is

$$\ell = m/v = \sqrt{(1-\theta)/v} > 0.$$

Set $U = W/\ell$ and $Y = B_0 + U$. Then

$$X = \ell(Y - p), \qquad \mathbb{E}[U \mid B_0] = 0. \tag{15}$$

Let U_0, U_1 have the two conditional laws, and put

$$w_i = \mathbb{E}U_i^2, \quad \mu_i = \mathbb{E}U_i^3, \quad s = qw_0 + pw_1 = \frac{v\theta}{1-\theta}, \quad \lambda = ns, \quad \ell = (v+s)^{-1/2}.$$

The moments μ_i are finite because $\rho < \infty$. Higher moments are used only for extremal laws, which have compact support by Proposition 2.1.

If $s = 0$, both nonnegative conditional variances vanish, so $U_0 = U_1 = 0$ almost surely. This is exactly the standardized Bernoulli law. If $s > 0$, the parameter $\alpha = pw_1/s$ lies in $[0, 1]$ and gives $w_0 = s(1-\alpha)/q$, $w_1 = s\alpha/p$.

The signs of X are still those defining the two classes, so expansion of the actual conditional moments gives

$$\begin{aligned}
\rho/\ell^3 &= v(1-2v) + 3v(w_0 + w_1) - q\mu_0 + p\mu_1, \\
\gamma/\ell^3 &= v(q-p) + 3v(w_1 - w_0) + q\mu_0 + p\mu_1, \\
\beta/\ell^2 &= v(q-p) + pw_1 - qw_0.
\end{aligned} \tag{16}$$

For example, on the nonpositive class $|X|^3 = \ell^3(p - U_0)^3$; on the positive class it is $\ell^3(q + U_1)^3$. Conditional centering removes the linear terms and gives the first identity; the other two follow in the same way. The three identities share the same pair (μ_0, μ_1).

3.2 A nonnegative moment defect

The constant $3 + \sqrt{10}$ has a simple geometric explanation. The signed third moment and the first absolute moment satisfy a circle inequality; the lengths of the support gaps are controlled by the same first absolute moment. Neither fact requires bounded support.

Define

$$\tau(t) = \mathbb{E}[X\mathbf{1}_{\{X \geq t\}}], \qquad \sigma = \sqrt{10}.$$

For $t < 0$, centering gives $\tau(t) = -\mathbb{E}[X\mathbf{1}_{\{X<t\}}] \geq 0$; for $t \geq 0$ nonnegativity is immediate. Signwise Tonelli integration gives

$$\int_{\mathbb{R}} \tau(t)\, dt = 1, \qquad 2\int_{\mathbb{R}} |t|\tau(t)\, dt = \rho. \tag{17}$$

On each bounded complementary interval (a,b) of the closed support, τ is constant in the interior, with value $\tau_{a,b}$. Put

$$J=\sum_{(a,b)}\tau_{a,b}(b-a)^2,\qquad D=(3+\sigma)\rho-\gamma-3J. \tag{18}$$

There are at most countably many gaps; the next proof shows that the nonnegative sum is finite. Endpoint atoms do not affect its integrals.

Lemma 3.1 (A moment circle and the gap inequality)**.** *For every $P\in\mathcal{P}$,*

$$\gamma^2+(2m)^2\le\rho^2,\qquad J\le\rho+2m,\qquad \gamma+tJ\le\bigl(t+\sqrt{1+t^2}\bigr)\rho\quad(t>0). \tag{19}$$

The last constant is sharp. Equality holds exactly at the standardized Bernoulli law with positive mass $p_t=(1+t-\sqrt{1+t^2})/2$. In particular, $D\ge 0$. If zero belongs to the support, then $D\ge(\sigma-1)\rho$; hence $D<(\sigma-1)\rho$ implies a central support gap $(-A,B)$ with $A,B>0$.

Proof. Write $a_\pm=\mathbb{E}X_\pm^2$ and $b_\pm=\mathbb{E}X_\pm^3$, so $a_++a_-=1$, $b_++b_-=\rho$ and $b_+-b_-=\gamma$. Cauchy–Schwarz on each sign class gives $a_\pm^2\le mb_\pm$. It also gives $m^2\le pa_+$ and $m^2\le qa_-$, where an atom at zero is included in q. Thus $m^2(1/a_++1/a_-)\le 1$, or $m^2\le a_+a_-$. Consequently $b_+b_-\ge(a_+a_-)^2/m^2\ge m^2$, proving the circle inequality since $\rho^2-\gamma^2=4b_+b_-$.

Every gap on one side of zero satisfies $(b-a)^2\le 2\int_a^b|x|\,dx$. If a central gap $(-A,B)$ exists, its weight is m and its excess over this integral is $2mAB$. Moreover $0\le\mathbb{E}[(X+A)(X-B)]=1-AB$. Summing the gap bounds by Tonelli proves $J\le\rho+2m$. If zero is in the support there is no central gap, so $J\le\rho$. Finally Cauchy–Schwarz in $\mathbb{R}^2$ gives $\gamma+2tm\le\sqrt{1+t^2}\rho$, proving the family and the assertions about D.

Equality in the circle forces equality in the two signwise Cauchy bounds: the positive and negative parts are each concentrated at one nonzero value. Equality in $m^2\le a_+a_-$ excludes mass at zero. For such a law, with positive mass p and $q=1-p$, one has $J=\rho+2m$. Equality in the last Cauchy bound requires $\gamma=2m/t$, equivalently $t(1-2p)=2p(1-p)$. Its unique solution in $(0,1)$ is the stated p_t. □

Corollary 3.2 (The zero-defect law)**.** *For $P\in\mathcal{P}$, $D=0$ if and only if*

$$X=(B_0-p_*)/\sqrt{p_*q_*},\qquad p_*=(4-\sqrt{10})/2,\quad q_*=1-p_*,\quad B_0\sim\text{Bernoulli}(p_*).$$

Proof. Apply the equality case of Lemma 3.1 with $t=3$. □

In particular the defect separates two losses:

$$D=\underbrace{\sigma\rho-\gamma-6m}_{\text{moment loss}}+3\underbrace{(\rho+2m-J)}_{\text{gap loss}},$$

each nonnegative. For quantitative estimates we need more information than their signs. Suppose a central gap $(-A,B)$ has been established, and write

$$T_+=\mathbb{E}X_+^3-mB^2,\quad T_-=\mathbb{E}X_-^3-mA^2,\qquad R_+=\mathbb{E}X_+^2-mB,\quad R_-=\mathbb{E}X_-^2-mA.$$

These four quantities are nonnegative. For the outer gaps put $J_o=J-m(A+B)^2$. A positive outer gap satisfies $(b-a)^2\le 2\int_a^b(t-B)\,dt$, so Tonelli gives

$$J_o^+\le 2\int_B^\infty(t-B)\tau(t)\,dt=\mathbb{E}[X_+(X_+-B)^2]=T_+-2BR_+\le T_+.$$

Similarly $J_o^- \le T_- - 2AR_- \le T_-$. Expanding D, using $\sqrt{\sigma+1}\sqrt{\sigma-1} = 3$, gives

$$D = m\{\sqrt{\sigma+1}A - \sqrt{\sigma-1}B\}^2 + (\sigma-1)T_+ + (\sigma+1)T_- \\ + 3(T_+ + T_- - J_o). \tag{20}$$

Every term is nonnegative. Cauchy–Schwarz also gives $1/(2\rho) \le m \le 1/2$: indeed $1 \le \rho\mathbb{E}|X| = 2m\rho$ and $\mathbb{E}|X| \le 1$. Since $\sigma - 1 > 2$, the decomposition yields $T_+ + T_- \le D/2$. If $D < 2$, then

$$m(A^2 + B^2) = \rho - T_+ - T_- \ge 1 - D/2, \qquad A + B \ge \sqrt{2 - D}.$$

3.3 One joint budget for endpoint motion and noise

Let $\bar{A} = m/q$, $\bar{B} = m/p$ be the conditional mean magnitudes, and let $\Delta_A = \bar{A} - A$, $\Delta_B = \bar{B} - B$. They are nonnegative.

Lemma 3.3 (Simultaneous gap inequality)**.** *For this central gap,*

$$D \ge \frac{(6+2\sigma)(1-\theta)^{3/2}}{\sqrt{pq}}(p - p_*)^2 + 6m\{(A+B)(\Delta_A + \Delta_B) + \Delta_A\Delta_B\} \\ + 3(\sigma+1)B\theta_+ + 3(\sigma+3)A\theta_-. \tag{21}$$

Proof. The stronger outer-gap bound in the preceding proof gives

$$D \ge (\sigma-1)\mathbb{E}X_+^3 + (\sigma+1)\mathbb{E}X_-^3 - 6mAB + 6BR_+ + 6AR_-.$$

Conditional cubic expansion gives

$$\mathbb{E}X_+^3 = m\bar{B}^2 + \mathbb{E}[(X + 2\bar{B})(X - \bar{B})^2; X > 0] \ge m\bar{B}^2 + 3B\theta_+,$$

and similarly $\mathbb{E}X_-^3 \ge m\bar{A}^2 + 3A\theta_-$. Moreover $R_+ = \theta_+ + m\Delta_B$ and $R_- = \theta_- + m\Delta_A$. Substitution produces the last two terms of (21) and the deterministic expression

$$m\{(\sigma-1)\bar{B}^2 + (\sigma+1)\bar{A}^2 - 6\bar{A}\bar{B}\} + 6m\{\bar{A}\bar{B} - AB + B\Delta_B + A\Delta_A\}.$$

The second brace equals $(A+B)(\Delta_A + \Delta_B) + \Delta_A\Delta_B$. For the first, use $m^2 = (1-\theta)pq$ and the exact identity

$$(\sigma+1)p^2 + (\sigma-1)q^2 - 6pq = (6+2\sigma)(p - p_*)^2.$$

This proves the displayed inequality. □

In count coordinates the inward displacements are

$$a_0 = p - A/\ell = \Delta_A/\ell, \quad a_1 = q - B/\ell = \Delta_B/\ell, \quad d_Y = a_0 + a_1 = (\Delta_A + \Delta_B)/\ell.$$

Thus $U_0 \le a_0$ and $U_1 \ge -a_1$. The physical displacement $\Delta_A + \Delta_B$ and the count displacement d_Y differ by ℓ.

4 From fixed-target contact to a coarse acquisition

We now derive quantitative support and moment bounds, then explain the finite enclosure that propagates them. The selected calculation has two coarse quadratic passes followed by four signed-moment passes. Every improvement is used only after its hypotheses have been established for the same attaining law. The final output is an acquisition bound, not a local exclusion or a proof of the conjecture. Let P be a centered, unit-variance law, S_j a sum of j independent copies, and set

$$\rho = \mathbb{E}|X|^3, \quad \beta = \mathbb{E}[X|X|], \quad \gamma = \mathbb{E}X^3, \quad c = C_{\mathrm{E}} = \frac{3+\sqrt{10}}{6\sqrt{2\pi}}.$$

The threshold H always has the units of the raw sum. A fixed-target maximizer maximizes

$$a(P,H) = \sqrt{n}\{\mathbb{P}(S_n \le H) - \Phi(H/\sqrt{n})\} - c\rho.$$

We use Proposition 2.1 and its closed-support contact formula (4). In particular its closed-support contact equality, not ratio optimality, is the variational premise here. We consider a maximizer with $a \ge 0$. The actual normalized score is $c + a/\rho$; no upper bound on it is assumed.

4.1 The first support bound

Theorem 4.1 (Initial moment and support entry)**.** *Assume $n \ge 42$, $C_{n-1} \le c$, and the universal structural inequality*

$$\sqrt{n}\,\Delta_n(P) \le \frac{3031}{10000}\left(\rho + \frac{323}{500}\right) \tag{22}$$

for standardized laws with finite third moment. Then a fixed-target maximizer with $a \ge 0$ satisfies

$$1 \le \rho < \frac{7361}{4000} < \frac{1841}{1000}, \qquad |\beta| < \frac{321}{500}, \qquad \operatorname{supp} P \subset (-96/25, 96/25). \tag{23}$$

The theorem is uniform in all real H and in both signs of a support point. It requires neither C_{n-2} nor an excess bound of order $1/n$.

The structural inequality (22) is the external iid theorem of Shevtsova [16]. Only this affine bound is needed for the displayed moment radius; it is not a consequence of the finite arithmetic in this section.

Proof. Since $c\rho \le \sqrt{n}\Delta_n(P)$ and $c > 819/2000$, elementary division gives

$$\rho < \frac{(3031/10000)(323/500)}{819/2000 - 3031/10000} = \frac{7361}{4000}.$$

The other affine bound $0.3322(\rho + 0.429)$ alone would give the larger number $712569/386500$; it must not be used to claim $7361/4000$. Put $m = \mathbb{E}X_+ = \mathbb{E}X_- > 0$ and $a_\pm = \mathbb{E}X_\pm^2 = (1 \pm \beta)/2 > 0$. Cauchy on each sign class gives $m\rho \ge a_+^2 + a_-^2$ and $m^2 \le a_+ a_-$. Hence

$$\rho \ge \frac{1+\beta^2}{\sqrt{1-\beta^2}}.$$

This function increases with $|\beta|$. For $r = 1841/1000$, $b = 321/500$,

$$(1+b^2)^2 - r^2(1-b^2) = \frac{93288849}{50000000000} > 0,$$

which proves the second assertion without assuming bounded support.

Write $v = n-1$, $s = \sqrt{n/v}$, $F_v(x) = \Phi(x/\sqrt{v})$ and $e_n = n(s-1) - 1/2$. Closed-support contact, the upper predecessor bound $F_{n-1,P}(H-t) \le F_v(H-t) + c\rho/\sqrt{v}$, and $\sqrt{n}(F_{n,P}(H) - \Phi(H/\sqrt{n})) = c\rho + a$ give

$$0 \le H_n(H,t) + c\{-|t|^3 + \tfrac{3}{2}\rho t^2 + 3\beta t + \rho e_n\} - na \quad (t \in \operatorname{supp} P), \tag{24}$$

where exactly

$$\frac{H_n(H,t)}{n^{3/2}} = F_v(H-t) - F_{v+1}(H) + tF'_{v+1}(H) - \frac{t^2-1}{2}F''_{v+1}(H).$$

The extra $-na$ can be discarded in this upper bound. This is the precise reason the predecessor assumption is $C_{n-1} \le c$, not a comparison with the actual score. No continuity assumption on the distribution function at $H-t$ is inserted.

Rationalization gives

$$ne_n = \frac{s^2(2s+1)}{2(s+1)^2}, \qquad s < 81/80 =: S, \qquad ne_n < \frac{793881}{2073680} < \frac{48}{125}.$$

The first expression increases for $s \ge 1$, as its logarithmic derivative is $2/s + 2/(2s+1) - 2/(s+1) > 0$. Thus $e_n < 8/875$.

For completeness we rederive the Gaussian estimate used at the radius. The heat identity $\partial_v F_v = F''_v/2$ permits spatial expansion of $F_v(H-t)$ to order four, variance expansion of F_{v+1} to order two, and variance expansion of F'_{v+1}, F''_{v+1} to order one. The cubic coefficient is $(-t^3+3t)/6$, and the quartic coefficient is $(t^4 - 6t^2 + 3)/24$. Therefore, for $t = \pm u$, $z_v = H/\sqrt{v}$,

$$H_n(H, \pm u)/s^3 = \pm A(1 - z_v^2)\phi(z_v) + B\phi'''(z_v) + \mathcal{E}, \tag{25}$$

where $A = (u^3 - 3u)/6$, $H_4 = u^4 - 6u^2 + 3$ and $B = H_4/(24\sqrt{v})$. The four error bounds after division by s^3 are respectively

$$\frac{u^5\|\phi^{(4)}\|_\infty}{120v}, \quad \frac{\|\phi^{(5)}\|_\infty}{48v^{3/2}}, \quad \frac{u\|\phi^{(4)}\|_\infty}{8v}, \quad \frac{|u^2-1|\|\phi^{(5)}\|_\infty}{16v^{3/2}}.$$

Gaussian Fourier inversion gives norms at most $3\phi(0) < 6/5$ and $8/\pi < 51/20$. Since $v \ge 41$ and $\sqrt{v} > 6$, for $u \ge 1$ this gives

$$|\mathcal{E}| < E(u) := u^5/4100 + 3u/820 + 17(3u^2-2)/78720.$$

These errors are uniform in H; they use no higher moment of P.

If $A, B > 0$ and $B \le A/4$, the leading part of (25) is at most $\phi(0)(A + 9B^2/(4A))$ for both signs. Indeed, for the positive sign and $|z_v| \le 1$, it is at most $\phi(0)(A - Az_v^2 + 3B|z_v|)$, and complete the square. Outside that interval the first term is nonpositive. For the negative sign the first term is at most $2e^{-3/2}A\phi(0) < A\phi(0)/2$. In both cases use $\|\phi'''\|_\infty < 2\phi(0)$: on $[0, \sqrt{3}]$ the polynomial $3x - x^3$ is at most two, and beyond $\sqrt{3}$ use $x^3 e^{-x^2/2} \le 3\sqrt{3}e^{-3/2} < 2$.

For frozen r, b above define

$$Q(u) = -u^3 + \tfrac{3}{2}ru^2 + 3bu + (8/875)r,$$
$$P(u) = \tfrac{2}{5}S^3\left(A^2 + \frac{H_4^2}{256 \cdot 41}\right) + S^3E(u)A + \frac{819}{2000}Q(u)A.$$

At $R = 96/25 < 6$ we have $A > 0$, $0 < H_4 < 36A$, hence $B < A/4$. The sign $Q(R) < 0$ is checked before replacing c by its lower endpoint. The upper bound in (24), multiplied by A, is then strictly below $P(R)$, and exact arithmetic gives

$$-P(R) = \frac{2214258283536204102723}{10250000000000000000000} > \frac{1}{5}.$$

To cover each entire exterior ray, differentiation of the exact Gaussian expression gives

$$|\partial_t H_n(H,t)| < \tfrac{2}{5}(t^2/2 + 1/2 + e_n).$$

For verification put $y = (H-t)/\sqrt{n}$, $z = H/\sqrt{n}$; the derivative is $n\phi(z) + Ht\phi(z) - ns\phi(sy)$. The second-order Taylor error between $\phi(y)$ and $\phi(z)$ costs $\phi(0)t^2/2$, while $\partial_s[s\phi(sy)] = -\phi''(sy)$ costs $\phi(0)n(s-1)$. Define $L(u) = \frac{2}{5}(u^2/2 + 1/2 + 8/875) + (819/2000)Q'(u)$. At R, the six numbers Q, Q', Q'', L, L', L'' are strictly negative. Since $Q''' = -6$ and $L'' = 2/5 - 6(819/2000) < 0$, all their required signs persist for $u \geq R$. Thus both radial upper bounds decrease from a strictly negative boundary value. The contact inequality excludes every support point with $|t| \geq R$. □

4.2 Higher moments and an actual Fourier event

Higher moments are used only after Theorem 4.1. For $x = |X|$ the nonnegative measure $d\mu_0 = (x-1)^2(x+1/2)\,dP$ has mass $\rho - 1$. Direct expansion gives

$$M_4 - 1 = \int \frac{(x+1)^2}{x+1/2}\,d\mu_0,$$
$$M_5 - 1 = \int \frac{x^3 + 2x^2 + 3x + 3/2}{x+1/2}\,d\mu_0,$$

where $M_j = \mathbb{E}|X|^j$. Both ratios increase for $x \geq 0$: they equal $x + 3/2 + 1/[4(x+1/2)]$ and $x^2 + 3x/2 + 9/4 + 3/[8(x+1/2)]$, whose derivatives are nonnegative. Writing their values at $R = 96/25$ as $k_4(R), k_5(R)$, we obtain

$$M_4 \leq \min(R\rho, 1 + k_4(R)(\rho-1)), \quad M_5 \leq \min(RM_4, 1 + k_5(R)(\rho-1)), \quad M_6 \leq RM_5. \tag{26}$$

Also $\gamma^2 \leq M_4 - 1$, by expanding $\mathbb{E}(X^2 - 1 - \gamma X)^2 \geq 0$.

There is no need to assume a central support gap before the first Fourier bound. For independent copies let $E = \{(X - X')^2 \geq 9/25\}$ and let M be any valid fourth-moment ceiling. Then

$$\mathbb{P}(E) \geq \alpha := \frac{(2 - 9/25)^2}{2M + 6} > 0, \quad |X - X'| \geq 3/5 \text{ on } E, \quad \mathbb{E}[(X - X')^2 \mid E] \leq 2/\alpha. \tag{27}$$

Indeed $\mathbb{E}[(X-X')^2; E] \geq 2 - 9/25$ and $\mathbb{E}(X-X')^4 = 2M_4 + 6 \leq 2M + 6$, so Cauchy proves the mass bound. The conditional second-moment bound follows from its unconditional value two. This is an event under one actual iid marginal, not a replacement characteristic function.

The same support proof will be used again on a smaller closed moment cell. Freeze upper bounds r, b for $\rho, |\beta|$ in its polynomial Q, and test a proposed radius before changing any moment bound. For $R \in [5/2, 96/25]$, the gates $A > 0$, $0 < H_4 < 36A$, $Q(R) < 0$, $P(R) < 0$, and the six negative ray signs suffice. Their strictness excludes the boundary atoms as well as the open exterior rays. A radius that fails a gate is not used. The selected finite search is $R = 2.50, 2.51, \ldots, 3.84$; it only chooses among proved implications.

4.3 The Fourier enclosure used by every pass

Let $f(t) = \mathbb{E}e^{itX}$ and $e_n(t) = e^{-nt^2/2}(1 - in\gamma t^3/6)$. For the remainder in (7), the compensated inversion proved in Appendix A gives

$$|E_n| \leq \frac{2\sqrt{n}}{\pi}\left\{\int_0^{1/2} \frac{|\Im[e^{-itH}(f(t)^n - e_n(t))]|}{t^3}dt + \int_{1/2}^{\infty} \frac{|f(t)^n - e_n(t)|}{t^3}dt\right\}.$$

The comparator is signed, has the correct mass and first moment, and its entire high-frequency tail is included. This formula applies at atoms and for arbitrary real H.

For clarity, the exact input and output of the selected Fourier step are as follows. Acquire $|X| \le R$ first, and choose common ceilings $M \le 11.1$, $g \le 1.85$, M_6 for $M_4, |\gamma|, \mathbb{E}X^6$, together with

$$|(3 - M_4)/24| \le A_4, \qquad |(10\gamma - \mathbb{E}X^5)/120| \le Z_5.$$

Put

$$\begin{aligned} E_6 &= M_6/720 + g^2/72 + (1/2 + g^2/144)^3/5, \\ B_6 &= A_4 + Z_5/2 + E_6/4, \\ a_0 = (N-1)/(2N), \quad b_0 &= (45 - M)(1 - 2/N)/96, \\ c_0 \ge \sqrt{\pi/2}(N/(N-1))^{3/2}, \qquad c_1 &\ge c_0/2. \end{aligned}$$

A complete admissible low-frequency triple is

$$\begin{aligned} L_0 &= c_0 Z_5 + c_1 A_4 g + 251 g^3/17280, \\ L_H &= c_0 A_4 + 251 g^2/4800, \\ L_2 &= E_6/(2a_0^2) + B_6^2/(2b_0^3). \end{aligned}$$

One may take $A_4 = \max(2, M-3)/24$ and $M_6 = RM_5$. The sharper fifth-order coefficient used by the finite cover is proved in Appendix A; it follows from the exact identity for $X^2 - \gamma X - 1$, not from an assumed fifth moment.

The high-frequency input is an actual event for two iid copies:

$$\mathbb{P}(\mathcal{E}) \ge \alpha > 0, \qquad |X - X'| \ge d_E > 0 \text{ on } \mathcal{E}, \qquad \mathbb{E}[(X - X')^2 \mid \mathcal{E}] \le M_E.$$

Proposition A.3 supplies a complete six-term bound K_H at $N = 42$, subject to its explicit positive damping and all-index gates. The first event is (27); an opposite-sign event is used only after an actual central gap is acquired below. Different valid windows may be minimized only after all six terms, including the comparator, are present. No characteristic function of an artificial binary law is substituted for f.

The exact scalar output is

$$|E_n| \le K_F + s_F|H|, \qquad K_F = \frac{16}{25}(L_0 + K_H + L_2/\sqrt{N}), \quad s_F = \frac{16}{25}L_H. \tag{28}$$

All square roots and exponential bounds in the finite evaluation are directed outward. The proof in the appendix applies to every actual $n \ge N$ and allows the law and threshold to vary with n. Thus no fixed-law limit is being used as a uniform bound.

4.4 Coarse coefficients before a central gap is known

Write the non-Fourier constant charge in (7) as

$$A_c^{\mathrm{law}} = \frac{3}{2}\beta(J + \gamma) + 3(\rho J_1 - J_2) + \frac{\rho}{2}M_4 - \frac{1}{4}M_5 - \frac{\rho}{2}.$$

Here A_c^{law} does not depend on c; the subscript records its role as the coefficient of c. We will bound it from above, retaining all signs in the later passes.

The following gap inequalities need no gap around zero:

$$-M_4/6 \le J_1 \le M_4/2, \qquad -3J_2 - M_5/4 \le 0, \qquad J \le \rho + 1. \tag{29}$$

For a gap (u, v) of length h, the first pair follows from

$$\int_u^v (t-u)t\,dt + \frac{1}{2}\int_u^v t^2 dt = hv^2/2, \quad \frac{3}{2}\int_u^v t^2 dt - \int_u^v (t-u)t\,dt = h(2u+v)^2/6.$$

Sum with the nonnegative gap weights and use $\int t^2\tau = M_4/3$. On a negative gap, reflection and

$$\frac{1}{3}\int_u^v x^3 dx - \int_u^v (v-x)x^2 dx = u^3(v-u)/3 \geq 0 \quad (0 \leq u \leq v)$$

prove the second assertion; the positive portion only improves it. The last bound follows from $J \leq \rho + 2m$ and $2m \leq 1$.

For $1 \leq r \leq \rho$ and $R \leq 6$, choose $\Gamma(R, r) \geq 0$ with

$$\Gamma(R,r) \geq \max_{r \leq v \leq R} \left\{ -\frac{v^2}{4} + \frac{(R+r)v}{3} - \frac{Rr}{2} + \frac{r^3(2R-r)}{12v^2} \right\}_+ . \tag{30}$$

Then

$$3(\rho J_1 - J_2) \leq \frac{3}{2}\Gamma(R,r)(\rho-\gamma) + \frac{3}{8}\rho^2 J. \tag{31}$$

On the negative portion of a gap discard $|t| < \rho$, where its charge is nonpositive. Extending the remaining positive integral gives $\int_r^v (R-x)x(x-r)\,dx$. Its quotient by v^2 is the expression in (30). Signwise Tonelli charges its upper bound against $\mathbb{E}X_-^3 = (\rho-\gamma)/2$. On the positive portion, $t(\rho - t) \leq \rho^2/4$ and $\sum \tau_{u,v}\int_u^v (t-u)dt = J/2$ give the other term. This argument includes a gap crossing zero.

The finite bound for Γ partitions $[r, R]$ into 64 closed intervals. On each, maximize the concave quadratic at its clipped vertex $2(R+r)/3$ and its nonnegative reciprocal term at the left endpoint. The alternative ceiling $5/2$ follows at $r = 1, R = 6$ from $3v^4 - 28v^3 + 66v^2 - 11 > 0$ on $[1, 6]$: the endpoint values are $30, 205$ and the two interior critical values are $(85 \mp 35\sqrt{5})/2 > 0$. The positive integral representation is increasing in R and decreasing in r, so that endpoint choice encloses the present domain.

The elementary square

$$V = \mathbb{E}[|X|^3(|X| - \rho)^2] = M_5 - 2\rho M_4 + \rho^3 \geq 0$$

gives $\rho M_4/2 - M_5/4 \leq \rho^3/4$. Thus on $\rho \in [l, u]$, $|\beta| \leq b$, $M_4 \leq M$, and with $0 \leq \rho - \gamma \leq \Delta$, two complete constant bounds are

$$\begin{aligned} A_1(\Delta) &= \tfrac{3}{2}b(2u+1) + \tfrac{3}{2}\Gamma(R,l)\Delta + \tfrac{3}{8}u^2(u+1) + u^3/4 - l/2, \\ A_2 &= \tfrac{3}{2}b(2u+1) + 2uM - l/2. \end{aligned} \tag{32}$$

Use $A = \min(A_1, A_2)$. The first uses (31); the second uses (29). They are alternative complete bounds, not partial costs to be mixed.

For the linear term K_1, the following three bounds are valid:

$$\begin{aligned} K_{\text{rad}} &= 3ub + \frac{R(R+2-3l/2)}{R+1/2}(u-1), \\ K_{\text{univ}} &= 6u - 3u^2/2 + 3b, \\ K_R &= Ru/2 + \max_{r \in \{l,u\}} |R/2 - 3r/2|g + 3b. \end{aligned}$$

For the first use the positive measure $d\mu_0 = (|X| - 1)^2(|X| + 1/2)dP$, of mass $\rho - 1$, and the exact identity

$$K_1 + 3\rho\beta = \int \operatorname{sgn}(X)\frac{x(x+2-3\rho/2)}{x+1/2}\,d\mu_0, \qquad x = |X|.$$

The ratio is convex in $x \geq 0$ and decreases in ρ. Its negative part is at most $31/40$ on $\rho \leq 1.85$; require its displayed endpoint ceiling to exceed that number. The second bound uses $|\mathbb{E}[X|X|^3]-3\gamma| \leq 3\rho$ for $R \leq 6$, and $\rho \leq 2$. For the third use the signwise fourth-moment bounds $0 \leq \mathbb{E}X_{\pm}^4 \leq R(\rho \pm \gamma)/2$. Finally (29) gives

$$|\varphi\Omega| \leq v_H := \frac{2}{5}\max\{1/2, 2M/3 - 1/2\}, \qquad \Omega = J_1 + (M_4 - 3)/6.$$

Lemma 4.2 (A stronger signed fourth-moment charge)**.** *Let $|X| \leq R$, $\rho \in [l, u]$, $|\beta| \leq b < 1$. Put $h_* = 1 + b$ and $a(r) = (3r - R)/2$. A valid bound for $|K_1|$ is the maximum on $[l, u]$ of*

$$B_R(r, b) = Rr - \tfrac{3}{2}r^2 + 3b + \tfrac{1}{2}(1 + b)a(r)^2.$$

If the three gates

$$h_* \geq a(u), \qquad 3 + 2a(l)h_* - \tfrac{3}{2}h_*^2 \geq 0, \qquad R - 3u + \tfrac{3}{2}h_*^2 \geq 0$$

hold, it may be intersected with the smaller bound

$$Ru - \tfrac{3}{2}u^2 - 3 + 3h_* + a(u)h_*^2 - h_*^3/2.$$

Proof. For $\epsilon = \pm 1$ set $h = 1 - \epsilon\beta > 0$ and $\delta = \rho - \epsilon\gamma$. Signwise Cauchy and $m \leq 1/2$ give $\delta \geq h^2$. The fourth moments on the two signs give

$$\epsilon K_1 \leq R\rho - \tfrac{3}{2}\rho^2 - 3 + 3h + a(\rho)\delta - \delta^2/(2h).$$

Completing the square in unrestricted δ and using $h \leq h_*$ proves the first bound. With $\delta \geq h^2$, the last three terms maximize to

$$\psi_a(h) = \begin{cases} h(3 + a^2/2), & 0 < h \leq a, \\ 3h + ah^2 - h^3/2, & h \geq a. \end{cases}$$

The first branch is increasing. The second has derivative $3+2ah-3h^2/2$, a concave quadratic whose endpoint values on $[0, h_*]$ are nonnegative under the second gate. The branches agree at $h = a$; the first gate places h_* in the second branch. The remaining derivative in ρ is $R-3\rho+3h_*^2/2$, nonnegative by the third gate. This proves the claim, including all endpoints. The quadratic maximum in the first bound is evaluated at its endpoints and, when present, its concave vertex. □

Take the minimum of the applicable bounds for $|K_1|$ and call it K. No assertion about the signs of $\beta\gamma$ or $K_1\gamma$ is used.

4.5 Two quadratic passes and the first actual gap

Use the 24 closed cells with endpoints

$$r_i = 1 + 3361i/96000, \quad i = 0, \ldots, 24, \qquad N = 42,$$

and directed constants

$$c_- = 0.4097321837023 < C_{\mathrm{E}} < c_+ = 0.4097321837025,$$
$$\phi_l = 997/2500 < \phi(0), \quad \sigma_l = 3.162277660 < \sqrt{10}.$$

The Gaussian constants follow from the rational Machin estimates in Lemma B.1 and Appendix A; the tighter c enclosure is checked by the same alternating expansions and rational squares. The interval encloses the fixed coefficient c, not the actual normalized score.

On a cell $[l,u]$, form the moments and the universal actual event above, and compute the complete Fourier pair K_F, s_F. In pass 1 use $\Delta = 2u$, $A = \min(A_1(2u), A_2)$, and, for $t \in [c_-, c_+]$,

$$f(t) = tK + v_H + s_F, \qquad q(t) = tl - g/5.$$

Require $q > 0$. The bound $\mathcal{R}_c \le K_F + cA + f(c)|H| - q(c)H^2$ is affine in c at each fixed $|H|$, so its maximum is at an endpoint before completing squares. Let an overline denote upward rounding to 10^{-10}. Compute

$$\begin{aligned} T_1 &= \overline{\max_{t=c_-,c_+} \{K_F + tA + f(t)^2/(4q(t))\}}, \\ \eta_1 &= (c_- l - T_1/N)/2, \\ U_1 &= \min\left\{T_1, \overline{\max_{t=c_-,c_+} \{K_F + tA + f(t)^2/[4(q(t)+\eta_1)]\}}\right\}, \\ d_1 &= \overline{6U_1/(N\phi_l)}. \end{aligned} \tag{33}$$

The strict gate $0 \le T_1 < Nc_-l$ is checked before η_1 or the resource bound is used. Corollary 2.4 then proves $\mathcal{R}_c \le T_1$, $\mathcal{U} \le U_1$, and $nD \le Nd_1$. The scalar bounds hold at every actual $n \ge N$.

For the second pass the circle in Lemma 3.1 gives

$$D \ge \sigma\rho - \gamma - 3\sqrt{\rho^2 - \gamma^2}, \qquad \sigma = \sqrt{10}.$$

Put $a_0 = \sigma_l - d_1/l$. A valid lower bound for γ/ρ is

$$Y(a_0) = \begin{cases} -1, & a_0 \le -1, \\ (a_0 - 3\sqrt{10 - a_0^2})/10, & -1 < a_0 \le \sqrt{10}. \end{cases} \tag{34}$$

Indeed $y + 3\sqrt{1-y^2}$ increases from -1 to $\sqrt{10}$ on $[-1, 1/\sqrt{10}]$. Solving its level equation on that branch gives the smaller quadratic root in (34); it has $a_0 - y \ge 0$, so squaring introduced no false branch. Every y below it violates the required lower level. A downward rational bound y_l is obtained by rounding the square root upward. Thus

$$\begin{gathered} g_l = \max\{-g, \sigma_l l - 3 - d_1, \min(ly_l, uy_l)\} \le \gamma, \\ \Delta = \min\{2u, (1-\sigma_l)l + 3 + d_1, u - g_l\}, \qquad \phi_- = \phi_l(1 - T_1/(Nc_-l)) > 0, \\ A = \min\{A_1(2u), A_1(\Delta), A_2\}, \\ q(t) = \max\{tl - g/5, tl + \tfrac{1}{2}\min(\phi_- g_l, (2/5)g_l)\}. \end{gathered}$$

The first two bounds also use $2m \le 1$. Each value of $q(t)$ is affine in t, since its two alternatives have the same slope l. Repeat the raw bound first, intersect it with T_1, then price phase using this new raw bound, intersect the resource with U_1, and finally retain $d_2 \le d_1$. A resource bound is never substituted for a raw bound in the density or phase license.

Certificate condition 4.3 (The two coarse passes)**.** The preceding coefficient domains, Fourier gates, endpoint enclosures and strict raw gates hold on every one of the 24 closed cells. Outward evaluation gives the following ceilings, and in every second-pass cell $d_2 < (\sigma_l - 1)l$:

pass	$\max T$	$\max U$	$\max d$	strict gap gates
1	13.2818901598	10.9330610254	3.9164138937	23
2	11.5164967977	9.9790315907	3.5746638454	24.

The minimum second-pass gap margin is at least $526024602543/1600000000000 > 0$.

The stated condition is a finite exact arithmetic obligation. The named two-pass proof object records and rechecks these inequalities; it is kept distinct from the analytic implications just proved. Although its implementation retains additional previously valid minima, the fresh quadratic ceilings attain all the displayed minima in this cover, so no earlier resource-refinement schedule is a mathematical premise here.

If zero belonged to the closed support, every gap would be on one side of zero and $J \le \rho$. Hence $D \ge (\sigma-1)\rho$, contradicting the strict second-pass test. Centering and positive variance therefore give an actual central gap $(-A, B)$, with positive endpoints. This acquires an event, rather than supposing an artificial two-point kernel.

4.6 Signed moments and physical gap costs

Write $\mu_4 = M_4$, $\eta_5 = M_5$, $\sigma = \sqrt{10}$. Put $a_\pm = (1 \pm \beta)/2$, $b_\pm = (\rho \pm \gamma)/2$. Both signs have positive probability and positive first moment, since a centered variance-one law cannot be supported on one half-line. Thus $|\beta| < 1$ and the signwise denominators below are positive whenever used; degenerate correction denominators are handled separately. Cauchy gives $m^2 \le a_+ a_-$ and $b_\pm m \ge a_\pm^2$. Consequently

$$\frac{(1+\beta)^{3/2}}{\sqrt{1-\beta}} - \rho \le \gamma \le \rho - \frac{(1-\beta)^{3/2}}{\sqrt{1+\beta}}, \qquad m \ge \max\left\{\frac{1+\beta^2}{2\rho}, \frac{a_+^2}{b_+}, \frac{a_-^2}{b_-}\right\}.$$

These statements also hold with a zero atom. Projection of $X|X|$ onto the span of X and $|X|$ gives

$$\mu_4 \ge \rho^2 + \frac{(\gamma - \rho\beta)^2}{1-\beta^2}.$$

For completeness, this follows by minimizing the nonnegative expectation $\mathbb{E}(X|X| - uX - v|X|)^2$ over u, v; its Gram matrix has diagonal entries one and off-diagonal entry β, and is positive definite.

Two support-localizing moment matrices give the additional upper bounds as follows. If $D_\pm = Ra_\pm - b_\pm > 0$, then

$$2m + \sum_{\pm} \frac{(Rm - a_\pm)^2}{D_\pm} \le R, \qquad \mu_4 \le R\rho - \sum_{\pm} \frac{D_\pm^2}{Rm - a_\pm}.$$

Indeed the first follows by summing the determinant inequalities for

$$\mathbb{E}[(R - |X|)(1, |X|)^{\mathsf{T}}(1, |X|); \pm X > 0].$$

The second follows from the corresponding matrices with an additional factor $|X|$. One more factor $|X|$ yields, sign by sign,

$$\mathbb{E}[X_\pm^5] \le R\mathbb{E}[X_\pm^4] - \frac{(Rb_\pm - \mathbb{E}[X_\pm^4])^2}{D_\pm}.$$

Zero denominators are treated by dropping their nonnegative correction; they never imply exclusion without a separate contradictory numerator. The common m is retained throughout.

A second useful upper bound for m follows by integrating $(x-a)^2(R-x) \ge 0$ for $x = |X| \in [0, R]$. For every fixed $a > 0$,

$$2m \le \frac{R - \rho + 2a + Ra^2}{a(2R + a)}.$$

The cover uses only the eleven rational choices $a = 2/10, \dots, 12/10$.

Write $\delta = \gamma - \rho\beta$. Cauchy and a square completion on each sign give the signed constant bound

$$\frac{\rho}{2}\mu_4 - \frac{1}{4}\eta_5 \le \frac{\rho^3}{4} - \frac{(\rho+\gamma)(\delta_+)^2}{8(1+\beta)^2} - \frac{(\rho-\gamma)((-\delta)_+)^2}{8(1-\beta)^2}.$$

Indeed on either sign, write $a = \mathbb{E}X_s^2$, $b = \mathbb{E}X_s^3$, $c = \mathbb{E}X_s^4$, $f = \mathbb{E}X_s^5$. Cauchy gives $c \ge b^2/a$, $f \ge c^2/b$, and $\rho c/2 - f/4 \le \rho^2 b/4 - (c-\rho b)^2/(4b)$. Distance to the half-line $c \ge b^2/a$ proves the stated penalty on summing both signs. Independently, $\mu_4^2 \le \rho\eta_5$ and the projection bound above give

$$\frac{\rho}{2}\mu_4 - \frac{1}{4}\eta_5 \le \frac{\rho^3}{4} - \frac{(\gamma-\rho\beta)^4}{4\rho(1-\beta^2)^2}.$$

Each is a complete deduction from the same square. There is a third independent deduction which is often stronger. For $Y = |X|$ and $\delta = \gamma - \rho\beta$, put

$$V = \mathbb{E}[Y(Y^2 - \rho Y)^2] = \eta_5 - 2\rho\mu_4 + \rho^3.$$

Since $\delta = \mathbb{E}[Y \operatorname{sgn}(X)(Y^2 - \rho Y)]$, weighted Cauchy gives $\delta^2 \le 2mV$. Hence

$$\frac{\rho}{2}\mu_4 - \frac{1}{4}\eta_5 \le \frac{\rho^3}{4} - \frac{\delta^2}{8m}. \tag{35}$$

The maximum of these three separately valid losses can be subtracted. Their sum cannot. A zero atom contributes zero to the weighted measure.

An actual central gap without assuming $D < 2$

The physical decomposition (20) shows that $D < (\sigma-1)\rho$ implies an actual central gap. Use its same nonnegative terms, writing $T = T_+ + T_-$ and $Q = (\sqrt{\sigma+1}A - \sqrt{\sigma-1}B)^2$. Hence $T \le D/(\sigma-1)$. The additional inequality $J \le \rho + 2mAB$ gives $6mAB \ge (\sigma-1)\rho - D$. Consequently, if $\rho \ge l$, $D \le d < (\sigma_l - 1)l$ and $m \le m_u$,

$$(A+B)^2 \ge \frac{l - d/(\sigma_l - 1) + \{(\sigma_l-1)l - d\}/3}{m_u} > 0. \tag{36}$$

This is the distance lower bound used in the complete high-frequency event. It uses the actual moment m and the actual support endpoints.

Moreover $A \le a_-/m$, $B \le a_+/m$ and $J \le \rho + 2m$. The last bound gives the compulsory defect

$$D \ge \sigma\rho - \gamma - 6m.$$

The inequalities $T - J_o \ge 2BR_+ + 2AR_-$ give the useful additional endpoint constraint

$$D \ge (\sigma-1)b_+ + 6a_+B - (\sigma+5)mB^2.$$

If $D < (\sigma-1)b_+$, this downward quadratic forces B to be at least its positive root. The reflected constraint has a_-, b_-, A and the coefficients $\sigma+1, \sigma+7$. In a closed moment box, positive coefficients are rounded down, negative coefficients up, and the lower square-root endpoint is used in the resulting root bound.

The actual opposite-sign event has probability $2pq$. There is no atom at zero. Cauchy gives the necessary probability interval $2m_-^2/(1+\beta_+) \le p \le 1 - 2m_-^2/(1-\beta_-)$ on a moment cell. In particular, $2pq \ge (1+\beta^2)^2/(2\rho^2)$, using $m \ge (1+\beta^2)/(2\rho)$ and $pq \ge m^2$. The minimum of $p(1-p)$ on a closed interval is attained at an endpoint.

For independent copies one has exactly

$$\mathbb{E}[(X-X')^2 \mid XX'<0] = \frac{1+\beta(1-2p)+4m^2}{2pq} = \frac{1+4m^2+\beta}{2p} + \frac{1+4m^2-\beta}{2q}.$$

For an upper bound, use the upper m endpoint and both endpoints of p and β. The expression is convex in p and affine in β. This improves the former $2/\mathbb{P}(XX'<0)$ bound inside the complete high-frequency estimate. Both copies still have the same actual marginal.

Preserving central and outer integral costs

Let

$$P_1(A,B) = B^3/3 + AB^2/2 - A^3/6, \quad P_2(A,B) = B^4/4 + AB^3/3 - A^4/12, \quad G = \rho P_1 - P_2.$$

Then $J_{1,c} = mP_1$, $J_{2,c} = mP_2$. Since $\partial_B G = B(A+B)(\rho - B)$,

$$G(A,B;\rho) \le G(A,\rho;\rho) = \rho^4/12 + \rho^3 A/6 - \rho A^3/6 + A^4/12.$$

The last expression increases with A because its derivative is $(\rho - A)^2(\rho + 2A)/6$, and is convex in ρ because its second ρ derivative is $\rho^2 + \rho A$. Its closed-rectangle maximum is therefore bounded using the upper A endpoint and both ρ endpoints.

Positive outer gaps satisfy $(t-a)(\rho - t) \le (\rho - B)_+^2/4$. On negative outer gaps the positive part is bounded by $(R-\rho)^2/4$ after extracting $|t|$. Integration gives

$$3(\rho J_1 - J_2)_o \le \frac{3}{8}\{(\rho - B)_+^2 T_+ + (R-\rho)^2 T_-\}.$$

The available weighted outer budget is $D - mQ$, not the entire D. The cover intersects this budget with the individual bounds $T_+ \le b_{+,u} - m_l B_l^2$ and $T_- \le b_{-,u} - m_l A_l^2$.

For the signed phase term $\Omega = J_1 + (\mu_4 - 3)/6$, retain

$$-RT_-/2 \le J_{1,o} \le RT_+/2.$$

The central minimum of P_1 occurs at B_l and an A endpoint; its maximum occurs at B_u and A clipped to the interval at $A = B_u$. A nonnegative polynomial bound is multiplied by m_u for an upper bound and m_l for a lower bound; a negative one uses the opposite endpoint. For example, the binary endpoints $A = 3/2$, $B = 2/3$ make this maximum negative; the sign switch is necessary even when $D < 2$. The fourth-moment sign intervals also sharpen $K_1 = \mathbb{E}[X|X|^3] - 3\beta - 3\rho\gamma/2$. The complete K_1, Ω, φ rectangles are used in the signed expression $|cK_1 + \varphi\Omega|$, without replacing it by two separate absolute values.

A compulsory first moment before each signed evaluation.

The physical inequality $D \ge \sigma\rho - \gamma - 6m$ is especially useful inside a cell. If its prior ceiling is $D \le d$, and $\rho \ge l$, $\gamma \le \gamma_+$, then

$$m \ge m_* := \max\{0, (\sigma_l l - \gamma_+ - d)/6\}. \tag{37}$$

If $m_* > 1/2$, the cell is empty. Otherwise the common sign-moment inequality $4m^2 + \beta^2 \le 1$ forces $|\beta| \le \sqrt{1 - 4m_*^2}$. This restriction must be intersected with the beta interval for this gamma cell, not carried from a different gamma cell. Recompute all dependent sign-moment ranges after the intersection. These are necessary conditions on one law; no sufficiency or moment-realizability assertion is made.

For completeness, the event distance and conditional moment refinements fit the Fourier input exactly. On the acquired actual gap, use $\mathcal{E} = \{XX' < 0\}$, a downward square root of (36), and the larger of the two valid probability lower bounds

$$\alpha = \max\left\{\frac{(1+\min|\beta|^2)^2}{2u^2},\quad 2\min_{p=p_-,p_+} p(1-p)\right\}.$$

The conditional second-moment ceiling is the smaller of $2/\alpha$ and the corner maximum of the exact expression displayed above. Its numerators are positive, it increases with m, is affine in beta, and is convex in p, so these corner choices are valid. Each event option is inserted into all six HIGH terms. Taking the minimum with the complete universal-event bound preserves the same actual Fourier integral. None of these statements follows merely from declaring a moment rectangle feasible.

4.7 The signed enclosure and its ordered update

On one closed moment cell retain bounds for (ρ, β, γ, m), an already proved support radius R, an incoming raw ceiling T, and the weighted defect invariant $nD \le Nd$. Thus $D \le d$ for every actual $n \ge N$. Intersect all necessary moment conditions just proved, with directed endpoint operations. A positive numerator with a zero denominator proves incompatibility only through its original positive-semidefinite inequality; a zero-over-zero correction is dropped. No division is made before this check. Bisection of the monotone first-moment necessary inequality keeps its upper endpoint, so its rounding cannot remove a law.

Here is an explicit description of the coefficient enclosures. Let M be the resulting upper fourth moment and let $[J_-, J_+]$ contain J; one can use

$$J_- = \max\{0, ((3+\sigma_l)l - \gamma_+ - d)/3\},\quad J_+ = \min\{u + 2m_+, ((3+\sigma_u)u - \gamma_-)/3\},$$

with a rational $\sigma_u > \sqrt{10}$. For the beta term take the full corner maximum of $3\beta(J+\gamma)/2$. For the gap charge take the minimum of the complete coarse charge (31) and the central-plus-outer charge just proved. Write this bound as G_* and the beta-term maximum as B_*. Let $P_* \ge 0$ be any lower bound for the maximum of the three independent fifth-moment penalties. Then a valid complete constant bound is

$$A = \min\{B_* + G_* + u^3/4 - l/2 - P_*,\quad B_* + 2uM - l/2\}. \tag{38}$$

The fourth semimoment inequalities and the signed product rectangle for $\rho\gamma$ give an interval $[k_-, k_+]$ containing $K_1 = \mathbb{E}X_+^4 - \mathbb{E}X_-^4 - 3\beta - 3\rho\gamma/2$. The central polynomial mP_1 and the signed outer bounds give an interval $[\omega_-, \omega_+]$ containing $\Omega = J_1 + (M_4 - 3)/6$. Intersect them with every previous valid interval. When multiplying by a moment endpoint, the sign of the coefficient is retained. These constructions explain every coefficient in the next quadratic, without assuming that unrelated corner maxima are attained by one common law.

Require the incoming raw gate $0 \le T < Nc_-l$. By Corollary 2.4,

$$\varphi = \phi(H/\sqrt{n}) \in [\phi_-, 2/5], \qquad \phi_- = \phi_l(1 - T/(Nc_-l)) > 0.$$

For a formal coefficient $v \in [c_-, c_+]$ set

$$\begin{aligned} B_v &= K_F + vA, \\ F_v &= \max_{k\in\{k_-,k_+\},\ \omega\in\{\omega_-,\omega_+\},\ \psi\in\{\phi_-,2/5\}} |vk + \psi\omega| + s_F, \\ Q_v &= vl + \tfrac{1}{2}\min\{\phi_-\gamma_-, (2/5)\gamma_-\}. \end{aligned} \tag{39}$$

The actual raw remainder is at most $B_c + F_c|H| - Q_c H^2$. The maximum over the signed rectangle is valid by successive affine maximization, and the absolute value preserves convexity. For fixed $|H|$, this expression is convex in c: the constant and curvature parts are affine and F_c is convex. Thus the two endpoints c-,c+ suffice, even if A is negative. A negative lower gamma must be priced with the upper density endpoint; it cannot be reflected inside the contact law.

Lemma 4.4 (One ordered signed step)**.** *Suppose all preceding cell bounds hold and $Q_v > 0$ at both endpoints. Let upward rounding be applied to every proposed numerical ceiling. Set*

$$\begin{aligned} T' &= \min\{T, \max_{v=c_-,c_+} (B_v + F_v^2/(4Q_v))\}, \\ \eta &= (c_l - T'/N)/2, \\ U' &= \max_{v=c_-,c_+} (B_v + F_v^2/[4(Q_v + \eta)]). \end{aligned}$$

A negative T' or U' excludes the cell. Otherwise $\mathcal{R}_c \le T'$, $\mathcal{U} \le U'$, and the new weighted defect bound is

$$nD \le Nd', \qquad d' = \min\{d, 6U'/(N\phi_l)\}.$$

A valid raw-threshold ceiling is

$$|H| \le \max_{v=c_-,c_+} \frac{F_v + \sqrt{F_v^2 + 4(Q_v + \eta)\max(0, B_v)}}{2(Q_v + \eta)}. \tag{40}$$

Proof. Complete the two raw squares first. The incoming strict gate persists under $T' \le T$, and gives $\eta > 0$. Only now apply Corollary 2.4 and complete the two resource squares with curvature $Q_v + \eta$. A negative complete ceiling is incompatible with the nonnegative raw or resource expression. The resource contains $n\phi(0)D/6$, so the displayed weighted bound follows without setting the actual sample size equal to N. If the maximum of both resource quadratics at a phase were negative, no witness could have that phase. Replacing B by its positive part enlarges each positive root and gives (40). Previous valid resource or phase bounds may also be intersected, but an unlicensed resource bound cannot be fed back as a raw ceiling. □

4.8 Closed coverage and the selected acquisition theorem

For each of the original 24 rho cells, acquire a support radius with the independent heat/ray test before recomputing higher moments. Each signed pass then partitions a full necessary beta enclosure into eight closed intervals, each resulting necessary gamma enclosure into eight closed intervals, and each necessary common-m enclosure into two closed intervals. The original beta interval is restored for every gamma slice before applying (37). Discarding an outer interval counts all its descendants, and requires one of the proved necessary conditions to be inconsistent. In every processed rho cell the active and excluded counts sum to 128.

Each surviving actual law satisfies the cell update of Lemma 4.4. Pool upper bounds by maxima and lower bounds by minima before the next pass. This may enlarge the possible set, but cannot lose an actual witness. A rho cell disappears from later passes only when all of its descendants have been excluded. Degenerate closed intervals may duplicate boundary work; they do not leave holes.

For export, intersect each surviving necessary probability interval with all 100 closed panels $[j/100, (j+1)/100]$, $j = 0, \dots, 99$. A cell meeting a shared endpoint contributes to both panels. From $\theta = 1 - m^2/[p(1-p)]$ obtain a panel upper bound using its lower m endpoint and the maximum of $p(1-p)$ on the intersection, including $1/4$ if it crosses $1/2$. Pool this simultaneously with the raw, weighted defect, rho, beta, and phase bounds of the same cells.

Certificate condition 4.5 (The four signed passes)**.** Starting with Condition 4.3, all support, moment, actual-event, Fourier, denominator, signed-curvature and ordered raw-license gates hold on the complete four-pass cover just described. Its exact rational evaluation contains 11264 nominal signed boxes in total. The complete 100-panel export satisfies, on every nonempty panel,

$$\mathcal{R}_c \le 2.1995174, \qquad nD \le 32.20665378, \qquad D \le 0.76682509,$$
$$\rho \le 71527/48000, \qquad |\beta| \le 0.50549963, \qquad |H| \le 1.83605680.$$

Fourteen rho cells survive, and thirty probability panels are nonempty. The largest support radius among the surviving final rows is 157/50; thus the export also supplies

$$\operatorname{supp} P \subset (-157/50, 157/50). \tag{41}$$

This is the named fixed-target coarse proof object, with its own targets and complete incidence requirements. It has a fresh exact acceptance record; numerical acceptance remains distinct from the analytic proof that its leaves enclose all possible witnesses. The preceding statements supply that reduction and explicitly identify the finite predicates used. They do not change an older certificate's hypotheses or its target.

Theorem 4.6 (Selected fixed-target coarse entry)**.** *Let $n \ge 42$ be an actual index and let an attained fixed-target pair have $a \ge 0$. Assume the true sharp predecessor $C_{n-1} \le C_{\mathrm{E}}$ and the external structural inequality* (22)*. Under Conditions 4.3 and 4.5, the stated complete probability cover and all the displayed coarse bounds hold for that pair.*

Proof. Theorem 4.1 supplies the actual support before higher moments are used. Appendix A and the actual-event constructions prove the complete Fourier bounds on their specified domains. The coarse coefficient inequalities and the two ordered quadratic steps establish the first actual gap. Every signed cell thereafter uses the moment and physical-gap bounds of this same law, followed by Lemma 4.4. Induction over the four passes preserves the weighted invariants. The complete closed cover and outward pooling prove that every witness reaches an incident output panel with the stated bounds. □

The argument contains no upper bound for the witness's actual score, no scalar score recurrence, and no second deleted-score assumption. It uses only the simple quadratic phase price. The following sections replace downstream panel arguments by one acquired strip and two direct comparisons at $n \ge 700$. The internal entry cover remains intact. The independent finite base is a separate input.

5 One acquired strip

The entry calculation will be used only through its weighted defect and raw-threshold bounds. The former already forces the sign split close to Esseen's two-point law. Two applications of contact then bound the entire conditional support, without a further partition of the sign probability. Throughout this section set

$$\sigma = \sqrt{10}, \qquad p_* = (4-\sigma)/2, \qquad B_* = 32.20665378.$$

The law, support gaps and sign split are those of Section 3; in particular D is the actual gap defect, not a moment surrogate. Write the central gap, once acquired, as $(-A_-, A_+)$, so that its endpoints are not confused with the sign indicator $B_0 = \mathbf{1}_{\{X>0\}}$.

Proposition 5.1 (Direct geometry)**.** *Let X be compactly supported, centered and of variance one. If $n \ge 700$ and $nD \le B_*$, its closed support has a gap $(-A_-, A_+)$ across zero, with $A_-, A_+ > 0$. In the actual decomposition $X = \ell(B_0 + U - p)$, put*

$$a_0 = p - A_-/\ell, \quad a_1 = q - A_+/\ell, \quad d = a_0 + a_1,$$
$$\theta = \mathbb{E}\{X - \mathbb{E}(X \mid B_0)\}^2, \qquad \lambda = n\mathbb{E}U^2 = n\theta/\ell^2.$$

Then $a_0, a_1 \ge 0$, $\mathbb{E}(U \mid B_0) = 0$, $U_0 \le a_0$, $U_1 \ge -a_1$, and

$$\frac{3}{8} < p < \frac{93}{200}, \qquad \theta < .0036, \qquad d < .004, \qquad \ell > 1.996, \tag{42}$$
$$\rho < \frac{9}{8}, \qquad .066148 < \beta < .2527, \qquad \lambda < \frac{7}{20}. \tag{43}$$

These conclusions require no contact or predecessor hypothesis.

Proof. We have $D < D_0 := .047$, because $700D_0 > B_*$. If zero were in the closed support, Lemma 3.1 would give $D \ge (\sigma - 1)\rho \ge \sigma - 1 > D_0$. Thus the central gap exists; centering supplies both signs. Conditional means imply $a_j \ge 0$ and $0 \le d < 1$. Put $v = pq$ and $w = 1 - \theta = v\ell^2$, and define

$$Q(p) = (\sigma + 1)p^2 + (\sigma - 1)q^2 - 6pq = (6 + 2\sigma)(p - p_*)^2.$$

We first retain all the costs of the actual conditional noises. Let

$$\int f \, d\mu_\pm = \mathbb{E}\big[(X - \mathbb{E}(X \mid B_0))^2 \mathbf{1}_{\{\pm X > 0\}} f(|X|)\big].$$

Their masses sum to θ. The outer-gap estimate gives the common budget

$$D \ge \ell w Q + E_0 + \int k_+ \, d\mu_+ + \int k_- \, d\mu_-, \tag{44}$$

where

$$E_0 = 6\ell w\{d(1 - d) + a_0 a_1\},$$
$$k_+(x) = (\sigma - 1)(x + 2\ell q) + 6A_+, \qquad k_-(x) = (\sigma + 1)(x + 2\ell p) + 6A_-.$$

Indeed

$$J_o^+ \le 2\int_{A_+}^{\infty} (t - A_+)\tau(t) \, dt = \mathbb{E}[X_+(X_+ - A_+)^2],$$

and the negative outer gaps have the same expression in magnitudes. Add the central contribution $m(A_- + A_+)^2$ and use

$$\mathbb{E}X_+^3 = m(\ell q)^2 + \int (x + 2\ell q) \, d\mu_+, \qquad R_+ = \mu_+(\mathbb{R}) + m(\ell q - A_+),$$

with the negative counterparts. Expansion yields (44). This also proves the budget for countably many gaps by monotone convergence; endpoint atoms do not alter the integrals.

Hereafter use $3.162277660 < \sigma < 3.162277661$. The weaker gap estimate $J \le \rho + 2mA_-A_+$, together with $\rho \ge 1$, $\gamma \le \rho$ and $m \le 1/2$, gives

$$A_-A_+ \ge z := \frac{3.162277660 - 1 - D_0}{3} > 0, \qquad w \ge A_-A_+ \ge z.$$

The second inequality follows from $A_- \le \ell p$, $A_+ \le \ell q$. All costs in (44) are nonnegative, so

$$D \ge \frac{w^{3/2}Q}{\sqrt{v}} \ge 2z^{3/2}Q.$$

Let Q_- replace σ in its polynomial expression by the lower rational bound. The two exact tests

$$4z^3Q_-(.36)^2 > D_0^2, \qquad 4z^3Q_-(.48)^2 > D_0^2$$

exclude $p \le .36$ and $p \ge .48$, respectively: the squared distance to $p_* \in (.36, .48)$ is monotone outside that interval.

On $.36 < p < .48$, $\ell \ge 2\sqrt{z} > 1.67$, and the noise prices satisfy

$$k_+(x)/\ell \ge q\{(\sigma+5)z/w + 2(\sigma-1)\} > 5.2,$$
$$k_-(x)/\ell \ge p\{(\sigma+7)z/w + 2(\sigma+1)\} > 5.2.$$

For the first use $x \ge A_+$ and $A_+ \ge z/(\ell p)$; for the second use their negative counterparts. Since $w \le 1$, the constants follow by taking $q = .52$ and $p = .36$. Therefore $D \ge 1.67(5.2)\theta$, whence $\theta < .006$, $w > .994$ and $\ell > 1.99$. Also $A_- + A_+ \ge 2\sqrt{z}$ gives

$$d \le 1 - 2\sqrt{zv/w} < .2, \qquad v \ge .36(.64).$$

Evaluating the kernels at their lower support endpoints in (44) now yields

$$D \ge \ell\{wQ + 6wd(1-d) + \theta \min(C_+ - b_+d, C_- - b_-d)\}, \tag{45}$$

where

$$C_+ = 3(\sigma+1)q, \quad C_- = 3(\sigma+3)p, \quad b_+ = \sigma+5, \quad b_- = \sigma+7.$$

Each branch increases in d on the current domain, since $6(.994)(.6) - .006(3.162277661+7) > 0$; both noise prices are positive. The inward cost alone excludes $d \ge .004$, because

$$1.99(.994)6(.004)(.996) > .047.$$

After replacing d by zero, the derivative in θ of either branch $\sqrt{(1-\theta)/v}\{(1-\theta)Q + \theta C\}$ has the sign of $(2-3\theta)C - 3(1-\theta)Q$. It is positive: $C > 6.49$, $Q < .05$ and $\theta < .006$. Hence $D \ge Q/\sqrt{v}$. This ratio decreases on $[.36, 3/8]$, and

$$Q_-(3/8)^2 > D_0^2(3/8)(5/8)$$

excludes that interval. On $[93/200, .48]$ instead use $Q \ge Q(93/200) > D_0/2$ and $\sqrt{v} \le 1/2$. Thus $3/8 < p < 93/200$. Since

$$C \ge \min\{3(\sigma+1)(107/200),\ 3(\sigma+3)(3/8)\},$$

the bound $D \ge \ell\theta C$, with $\ell > 1.99$, improves $\theta < .0036$ and consequently $\ell > 1.996$.

The exact conditional cubic expansion gives

$$\rho = w^{3/2}\rho_B(p) + \int (x + 2\ell q)\, d\mu_+ + \int (x + 2\ell p)\, d\mu_-, \qquad \rho_B(p) = \frac{1-2v}{\sqrt{v}}.$$

Because the two integrands are priced by at least $\sigma - 1$ in (44),

$$\rho \le \rho_B(p) + D/(\sigma-1) < 1.1 + .047/(54/25) < 9/8.$$

Here $v \ge 15/64$ implies $\rho_B < 1.1$ by squaring positive quantities. The signed-second-moment identity gives

$$1 - 2p - 2q\theta \le \beta \le 1 - 2p + 2p\theta;$$

its endpoints imply $.066148 < \beta < .2527$.

To bound the scaled noise λ, retain the cubic cost rather than only its sign. If Z is centered, $Z \le a$, $a > 0$, and $t = \mathbb{E}Z^2$, then

$$0 \le \mathbb{E}[(a - Z)(Z + t/a)^2] = at - \mathbb{E}Z^3 - t^2/a.$$

Apply this to U_0 and $-U_1$. A zero endpoint forces the corresponding noise to vanish, so no division by zero is needed. For $\lambda_0 = nq\mathbb{E}U_0^2$, $\lambda_1 = np\mathbb{E}U_1^2$, the same gap-budget expansion, before bounding these two cubics, reads

$$\begin{aligned} nD/\ell^3 \ge\ & nvQ + 6nv\{d(1-d) + a_0a_1\} \\ & + \{3(\sigma+3)p - (\sigma+7)a_0\}\lambda_0 + \{3(\sigma+1)q - (\sigma+5)a_1\}\lambda_1 \\ & + \frac{(\sigma+1)\lambda_0^2}{nqa_0} + \frac{(\sigma-1)\lambda_1^2}{npa_1}. \end{aligned}$$

Terms with $a_j = 0$ are absent. Set $\bar{d} = .004$ and use $d(1-d) + a_0a_1 \ge (1-\bar{d})(a_0 + a_1)$. Pair each displacement cost with its reciprocal cubic cost by AM–GM; for example

$$6nv(1-\bar{d})a_0 + \frac{(\sigma+1)\lambda_0^2}{nqa_0} \ge 2\lambda_0\sqrt{6(1-\bar{d})(\sigma+1)p}.$$

Thus

$$nD \ge \ell^3\{nvQ + \lambda_0 Y_0(p) + \lambda_1 Y_1(p)\}, \tag{46}$$

where

$$\begin{aligned} Y_0 &= 3(\sigma+3)p - (\sigma+7)\bar{d} + 2\sqrt{6(1-\bar{d})(\sigma+1)p}, \\ Y_1 &= 3(\sigma+1)q - (\sigma+5)\bar{d} + 2\sqrt{6(1-\bar{d})(\sigma-1)q}. \end{aligned}$$

Each increases in its respective argument p or q. At $p = 3/8$ and $q = 107/200$, lower bounds 3 and 2.6 for the roots give $Y_0 > 12.8$, $Y_1 > 11.8$. Since $\lambda_0 + \lambda_1 = \lambda$,

$$B_* \ge nD > 1.996^3(11.8)\lambda \quad (\lambda > 0), \qquad 1.996^3(11.8)(.35) > B_*.$$

This proves $\lambda < .35$; at zero noise it is immediate. □

Proposition 5.2 (Two contact rays)**.** *Under Proposition 5.1, suppose (P, H) is an attained fixed-target pair at C_{E} with nonnegative excess and true predecessor $C_{n-1} \le C_{\mathrm{E}}$. Then*

$$-.6 < U_0 \le a_0, \qquad -a_1 \le U_1 < .6.$$

The support conclusion is uniform in the raw threshold H.

Proof. Use the contact upper expression (24). The all-threshold Gaussian heat estimate and its radial derivative bound from the proof of Theorem 4.1 give, for $t > \sqrt{3}$,

$$\begin{aligned} H_n(H, \pm t) \le h(t) &:= S^3\left\{\frac{2}{5}\left(\frac{t^3 - 3t}{6} + \frac{|t^4 - 6t^2 + 3|}{72}\right) + \frac{t^5}{4100} + \frac{3t}{820} + \frac{17(3t^2-2)}{78720}\right\}, \\ |\partial_t H_n(H, \pm t)| < g(t) &:= \frac{2}{5}(t^2/2 + 1/2 + \eta), \qquad S = 81/80, \quad \eta = 8/875. \end{aligned}$$

The first bound follows directly from (25): $\|\phi'''\|_\infty < 2\phi(0)$, $n - 1 \ge 41$, and $\sqrt{n-1} > 6$. No derivative of the absolute-value majorant is taken.

At a support point of magnitude t, contact requires a nonnegative upper expression $H_n(H, \pm t) + C_{\mathrm{E}} Q_b(t)$, where

$$Q_b(t) = -t^3 + \tfrac{3}{2} r t^2 + 3bt + \eta r, \quad r = 9/8, \quad b = \begin{cases} -.066148 & \text{negative ray,} \\ .2527 & \text{positive ray.} \end{cases}$$

These replacements follow with their signs from (43); the discarded excess term is nonpositive. Take $c_0 = 819/2000 < C_{\mathrm{E}}$ and the two fixed boundaries

$$t_- = 1.996(3/8 + .6) = 1.9461, \qquad t_+ = 1.996(107/200 + .6) = 2.26546.$$

For the respective b, direct rational substitution at each boundary gives

$$t^2 > 3, \qquad h(t) + c_0 Q_b(t) < 0, \qquad Q_b, Q_b', Q_b'' < 0, \qquad L, L', L'' < 0, \quad L = g + c_0 Q_b'.$$

The two boundary upper bounds are less than $-.40$ and $-.13$, respectively. These are just polynomial comparisons at two rational points. Since $Q_b''' = -6$ and $L'' = 2/5 - 6c_0 < 0$, the required signs persist along both outward rays. The derivative of the actual upper expression is below L, as $Q_b' < 0$ and $C_{\mathrm{E}} > c_0$. It therefore stays strictly negative beyond either boundary. Contact excludes every support point there, including boundary atoms.

On the negative class, $X > -t_-$ and $\ell > 1.996$ give $U_0 = X/\ell + p > -(3/8 + .6) + p > -.6$. On the positive class, $X < t_+$ gives $U_1 = X/\ell + p - 1 < 107/200 + .6 + p - 1 < .6$. The inward endpoints and conditional centering are the actual ones from Proposition 5.1. □

The pointwise normalization is worth recording once:

$$1 = v\ell^2 + \theta, \qquad s = \mathbb{E}U^2 = \theta/\ell^2, \qquad v + s = \ell^{-2}, \qquad X = \frac{B_0 + U - p}{\sqrt{v+s}}. \tag{47}$$

Thus the law to be compared in the next section is exactly the original law, and its threshold remains the raw H. Reflection, when required, selects the oriented positive witness before acquisition; neither its signed defect budget nor its probability interval is transported from the opposite orientation.

6 The direct comparison with a binomial law

The acquired strip admits a single comparison on its whole probability interval. Very small noise is controlled by the loss at one count and fourth moments. For the rest, five adjacent counts give a uniform crossing loss. Both arguments concern the actual conditional laws of one marginal.

Theorem 6.1 (The common-strip comparison)**.** *Let $n \geq 700$ be an integer, $B \sim \mathrm{Bernoulli}(p)$, and $\mathbb{E}(U \mid B) = 0$, where $p \in [3/8, 93/200]$. Write U_j for the conditional noise given $B = j$, and suppose*

$$-3/5 \leq U_0 \leq a_0, \qquad -a_1 \leq U_1 \leq 3/5, \qquad a_0, a_1 \geq 0, \quad a_0 + a_1 \leq 1/250.$$

Put $q = 1 - p$, $v = pq$, $s = \mathbb{E}U^2$, $\lambda = ns$, $X = (B + U - p)/\sqrt{v+s}$ and $\rho = \mathbb{E}|X|^3$. For independent copies and every raw threshold $|H| \leq 46/25$,

$$\sqrt{n}\left\{\mathbb{P}\left(\sum_{i=1}^{n} X_i \leq H\right) - \Phi(H/\sqrt{n})\right\} - C_{\mathrm{E}}\rho < -\lambda/100 \qquad (0 < \lambda \leq 7/20). \tag{48}$$

At $\lambda = 0$ the same excess is strictly negative.

The proof below uses only the displayed hypotheses. Contact and the sharp predecessor are needed earlier to acquire this strip, not for its local comparison. The small- and larger-noise intervals meet at $\lambda = 1/100$.

6.1 Three elementary ingredients

Write $\pi_j = \binom{n}{j}p^j q^{n-j}$, $F_B(j) = \sum_{i\le j}\pi_i$, and $\rho_B = (1-2v)/\sqrt{v}$. We use Schulz's strict finite-binomial theorem [13, Theorem 1] in the form

$$\sqrt{n}\{F_B(j) - \Phi((j-np)/\sqrt{nv})\} < C_{\mathrm{E}}\rho_B \quad (0<p<1, n<\infty). \tag{49}$$

Its strictness concerns every finite binomial law; it is not inferred from the limiting lower approximation for the universal constant.

The factorial estimates used throughout are finite Stirling bounds:

$$\log(j!) = (j+\tfrac{1}{2})\log j - j + \tfrac{1}{2}\log(2\pi) + r_j, \qquad 0 < r_j < 1/(12j).$$

For completeness, with $x = 1/(2j+1)$, $r_j - r_{j+1} = \operatorname{arctanh}(x)/x - 1 = \sum_{l\ge1} x^{2l}/(2l+1)$ lies strictly between 0 and $1/(12j(j+1))$. Wallis's product fixes the limiting constant, so $r_j \to 0$; summing these differences proves the bounds. One useful consequence is

$$\max_j \mathbb{P}\{\mathrm{Bin}(M,p) = j\} < \frac{41}{100\sqrt{Mv}} \quad (M \ge 43, p \in [1/3, 2/3]). \tag{50}$$

Indeed a mode has $z = j/M = p + \delta/M$, $|\delta| \le 2/3$. Since $v \ge 2/9$ and $|1-2p| \le 1/3$, $z(1-z) \ge v(1 - 1/M - 2/M^2) > 39v/40$. Stirling, the nonnegative relative entropy, and $e^{1/516} < 516/515$ bound the mass by $(2/5)\sqrt{40/39}(516/515)/\sqrt{Mv}$, which is smaller than the right side of (50) by a rational square comparison.

Finally the cubic correction has the favorable sign on the full strip. Let $s_j = \mathbb{E}U_j^2$, so $s = qs_0 + ps_1$, and set $r = 3/8$, $d = 1/250$, $v_* = 15/64$. The conditional supports have fixed signs relative to p, since $d < r$. Expansion of the raw third absolute moment gives

$$\mathbb{E}|B + U - p|^3 = v(1-2v) + 3v(s_0+s_1) - q\mathbb{E}U_0^3 + p\mathbb{E}U_1^3.$$

The inward bounds imply $\mathbb{E}U_0^3 \le a_0 s_0$, $\mathbb{E}U_1^3 \ge -a_1 s_1$. Together with $(1+x)^{3/2} \le 1 + 3x/2 + 3x^2/8$ for $x \ge 0$, this yields

$$\begin{aligned}(v+s)^{3/2}(\rho - \rho_B) &\ge s\left\{3r - d - \frac{3}{2}(1-2v) - \frac{3(1-2v)s}{8v}\right\} \\ &\ge s\left\{\frac{2593}{8000} - \frac{17}{20}s\right\} > 0, \qquad 0 < s \le \frac{1}{2000}.\end{aligned} \tag{51}$$

Here $s \le (7/20)/700 = 1/2000$, and the displayed coefficient is $3r - d - \frac{3}{2}(1-2v_*)$. Thus every nonnegative coefficient of $\rho_B - \rho$ can be discarded in an upper comparison.

6.2 Small noise: one count and the whole incoming tail

Assume $0 < \lambda \le L = 1/100$. Choose an integer k with

$$t = np + H\sqrt{v+s} = k + u, \qquad -3/10 \le u \le 7/10, \qquad \delta = k - np.$$

The phase interval has length one. It need not use the same anchor as the larger-noise proof. Let W_j have the law of the sum of j independent U_1 variables and $n-j$ independent U_0 variables. For $C \ge 0$, subtract the signed binomial excess at k from the actual excess:

$$\begin{aligned}\Delta_C = \sqrt{n}\left\{\mathbb{P}\left(\sum X_i \le H\right) - \Phi(H/\sqrt{n})\right\} - C\rho \\ - \sqrt{n}\{F_B(k) - \Phi(\delta/\sqrt{nv})\} + C\rho_B.\end{aligned}$$

Exact conditioning gives

$$\mathbb{P}\left(\sum X_i \le H\right) - F_B(k) = -\sum_{i=0}^{k} \pi_{k-i}\mathbb{P}(W_{k-i} > u+i) + \sum_{i=1}^{n-k} \pi_{k+i}\mathbb{P}(W_{k+i} \le u-i).$$

Outgoing events are strict and incoming events closed; this convention includes all threshold atoms. Define

$$f(u) = \frac{\Phi((\delta+u)/\sqrt{n(v+s)}) - \Phi(\delta/\sqrt{n(v+s)})}{\pi_k}, \qquad O = \mathbb{P}(W_k > u) + f(u),$$

$$T_{\text{in}} = \sum_{i=1}^{n-k} \mathbb{P}(W_{k+i} \le u-i), \qquad \mathcal{R} = \sqrt{n}\{\Phi(\delta/\sqrt{nv}) - \Phi(\delta/\sqrt{n(v+s)})\} + C(\rho_B - \rho).$$

Keeping the own-count loss and discarding the other nonpositive outgoing terms gives

$$\Delta_C \le -m_k(O - RT_{\text{in}}) + \mathcal{R}, \qquad m_k = \sqrt{n}\pi_k, \tag{52}$$

whenever $\pi_{k+i}/\pi_k \le R$ for all legal incoming counts.

Uniform normalization.

Set $D_* = 5/3$, $b_* = 1/20$, $T_* = 101/10000$ and $c_* = 2/525$. Because $v \ge v_*$ and $v + s < 1/4$, $|t - np| < 23/25$. Consequently $|\delta| < D_*$ globally and $|\delta| < 1$ when $|u| \le b_*$. Write $\mathcal{K}(z,p) = z\log(z/p) + (1-z)\log((1-z)/q)$ for binomial relative entropy; this is distinct from the physical gap defect D. The segment between p and k/n lies inside $[.37, .47]$; in particular $1 \le k \le n-1$. On this segment

$$z(1-z) > 23/100, \quad |\partial_z^3 \mathcal{K}(z,p)| < 5, \quad |(\log(z(1-z)))'| < 6/5, \quad (12z(1-z))^{-1} < 3/8.$$

Stirling and entropy Taylor expansion therefore give

$$m_k = \frac{\phi(0)}{\sqrt{v}} e^{-\delta^2/(2nv)+e}, \qquad n|e| < Z(D_0) := 3D_0/5 + 3/8 + D_0^3/840 \tag{53}$$

for $|\delta| \le D_0 \le D_*$. The three terms respectively bound the logarithmic prefactor, factorial remainder and entropy remainder. Also

$$\log f'(z) = -\tfrac{1}{2}\log(1+s/v) + \frac{\delta^2}{2nv} - \frac{(\delta+z)^2}{2n(v+s)} - e,$$

so, on an integration interval $|z| \le b_0$,

$$n|\log f'(z)| \le \frac{11}{5}\lambda + \frac{2D_0b_0 + b_0^2}{2v_*} + Z(D_0).$$

The coefficient $11/5$ exceeds $1/(2v_*) + D_*^2/(1400v_*^2)$. Using $e^{-x} \ge 1-x$, $e^x \le (1-x)^{-1}$ and $\phi(0) > 199/500$ now gives

	$\lvert u\rvert \le 1/20$	$-3/10 \le u \le 7/10$
m_k lower bound	3/4	3/4
f' lower bound	99/100	49/50
f' upper bound	101/100	51/50
R upper bound	51/50	6/5

(54)

The derivative estimates hold on the whole segment from 0 to u. For the global ratio, (50) gives $\sqrt{n}\max_j \pi_j < 17/20$, hence $R < (17/20)/(3/4) < 6/5$. In the inner interval the first ratio is

$$\frac{\pi_{k+1}}{\pi_k} = 1 + \frac{p-\delta-1}{nv+q(\delta+1)} < 1 + \frac{93/200}{700v_*} < 51/50;$$

every later ratio is at most one, since $p-\delta < 2$. By (51) the moment correction is nonpositive. Differentiating the normal CDF and using $\phi(0)/(2v_*^{3/2}) < 2$ therefore gives

$$\mathcal{R} < 2\lambda/n \quad (|u| \le b_*), \qquad \mathcal{R} < 4\lambda/n \quad \text{on the full phase interval.} \tag{55}$$

Indeed the positive displacement is bounded by 1 or $D_* < 2$.

No truncation of the incoming tail.

Since $|s_1 - s_0| \le (8/3)s$, the actual own-count variance $t_0 = \mathbb{E}W_k^2$ satisfies $.99\lambda < t_0 < 1.01\lambda \le T_*$. For each incoming count,

$$t_i = \mathbb{E}W_{k+i}^2 \le \lambda\{1 + c_*(D_0 + i)\}, \qquad \mathbb{E}W_{k+i}^4 \le A^2 t_i + 3t_i^2, \quad A = 3/5,$$

where $D_0 = 1$ in the inner interval and $D_0 = D_*$ otherwise. The moment bound follows from independence and $|U_j| \le A$. For $a = 1 + c_* D_0$ put

$$K(a) = A^2\left(\frac{13a}{12} + \frac{5c_*}{4}\right) + 3L\left(\frac{13a^2}{12} + \frac{5ac_*}{2} + \frac{5c_*^2}{3}\right).$$

Fourth-moment Markov at distance $i-u$, and extension of the nonnegative sum to all positive integers, give

$$\begin{aligned} T_{\text{in}} &< \frac{K(1+c_*)}{(1-b_*)^4}\lambda < .53\lambda \quad (|u| \le b_*), \\ T_{\text{in}} &< \frac{.43\lambda}{(1-u)^4} \qquad\qquad (0 \le u \le .7), \\ T_{\text{in}} &< .43\lambda \qquad\qquad (u \le 0). \end{aligned} \tag{56}$$

Here $K(1+c_*D_*) < .43$, and $\sum i^{-4} < 13/12$, $\sum i^{-3} < 5/4$, $\sum i^{-2} < 5/3$. These series bounds follow by retaining respectively five, three and five terms and estimating the remaining sum by an integral. For $u \ge 0$ use $i - u \ge (1-u)i$; for $u \le 0$ use $i - u \ge i$.

The own-count loss at the central phases.

With $\alpha = 13/25$ and $M = 5\alpha^4$, the quartic

$$h_\alpha(x) = \frac{-x^4 + 3\alpha^2 x^2 + 2\alpha^3 x}{M} \le \mathbf{1}_{(0,\infty)}(x)$$

is a strict-event minorant: its numerator factors as $-x(x+\alpha)^2(x-2\alpha)$, and its positive maximum divided by M is $(9+6\sqrt{3})/20 < 1$. Let $r = zt_0$ and set

$$F_{\varepsilon,b}(z,t) = \frac{3\alpha^2 - A^2 - 3t}{M} + \left(\frac{2\varepsilon}{5\alpha} + b - \frac{4At}{M}\right)z + \frac{3\alpha^2 t - 6t^2}{M}z^2 - \frac{t^3 z^4}{M}.$$

Expanding $h_\alpha(W_k + r)$ for $u = -r$, or $h_\alpha(W_k - r)$ for $u = r$, and using $|\mathbb{E}W_k^3| \le At_0$, $\mathbb{E}W_k^4 \le A^2 t_0 + 3t_0^2$, yields

$$O/t_0 \ge F_{1,-101/100}(z,t_0) \quad (u=-r), \qquad O/t_0 \ge F_{-1,99/100}(z,t_0) \quad (u=r).$$

Both functions exceed $3/5$ on $0 \le z \le 3/2$, $0 \le t \le T_*$. To check the whole rectangle, use concavity in t and check only $t = 0, T_*$. At those endpoints replace z^4 by $(9/4)z^2$ in its negative term. The negative-phase quadratic decreases on $[0, 3/2]$, and the positive-phase quadratic increases; their minima are respectively at $z = 3/2$ and $z = 0$. All four rational endpoint values exceed $3/5$.

For negative phases $3t_0/2 \le r \le b_*$, Cantelli gives

$$O \ge \frac{r^2}{t_0 + r^2} - \frac{101}{100} r > \frac{3}{5} t_0.$$

Its derivative at $r = 3t_0/2$ is positive and first increases, then decreases. The minimum is therefore at an endpoint. At the first use $t_0 \le T_*$ after division by t_0; at the other use $b_*^2/(T_* + b_*^2) - (101/100)b_* > (3/5)T_*$. For positive phases beyond this switch, $O \ge (99/100)r \ge (297/200)t_0 > (3/5)t_0$. The switch lies below b_* because $3T_*/2 < b_*$. Thus

$$O > \frac{3}{5} t_0 > .59\lambda \qquad (|u| \le b_*). \tag{57}$$

The value $u = 0$ is included, with $h_\alpha(0) = 0$.

The outer phases and the final margin.

For $u = -r$, $b_* \le r \le 3/10$, Cantelli gives

$$O \ge \frac{r^2}{T_* + r^2} - \frac{51}{50} r > 7/50 \ge 14\lambda.$$

The derivative is positive at b_* and has at most one downward zero; both endpoint values exceed $7/50$. For $b_* \le u \le 7/10$,

$$O - RT_{\text{in}} \ge \frac{49}{50} u - \frac{(6/5)(43/100)L}{(1-u)^4} > 1/25 \ge 4\lambda.$$

This function is concave and both endpoints exceed $1/25$. In the inner interval, (56) and (57) give $O - RT_{\text{in}} > (.59 - (51/50).53)\lambda$. Substitute these three margins and (55) into (52), using $m_k > 3/4$. The smallest final margin is the inner one, and

$$\frac{3}{4}\left(\frac{59}{100} - \frac{51}{50}\frac{53}{100}\right) - \frac{2}{700} > \frac{1}{100}.$$

Hence $\Delta_C < -\lambda/100$. Taking $C = C_{\text{E}}$ and applying (49) proves (48) for $0 < \lambda \le 1/100$.

6.3 Larger noise: five counts and four endpoint checks

Now assume $1/100 \le \lambda \le 7/20$ and choose $k = \lfloor t \rfloor$, $u = t - k \in [0, 1)$, $w = t - np$. The following constants will be used only in this subsection:

$$N = 700, \quad p_- = 3/8, \quad p_+ = 93/200,$$
$$r = 3/8, \quad q_* = 107/200,$$
$$v_- = 15/64, \quad v_+ = 9951/40000, \quad A = 3/5, \quad d = 1/250,$$
$$\eta = 2/5, \quad \beta_0 = 41/100, \quad \pi_+ = 22/7, \quad \phi_- = 199/500, \quad \phi_+ = 2/5,$$
$$\kappa = 3, \quad \xi = 93/100, \quad c_- = 6, \quad \zeta = 23/100, \quad m_* = 3/4, \quad V_* = 113/1000,$$
$$\bar{c} = 79/200, \quad j_0 = 1/5, \quad j_1 = 5/2.$$

In particular $|w| < \xi < 1$, since $(46/25)^2(v_+ + (7/20)/N) < \xi^2$. Also $N(r - d) > \xi$, so $na_0 \le t < n(1 - a_1)$.

An exact weighted crossing identity.

On an auxiliary space take independent copies of the conditional laws and write

$$T_j = j + \sum_{l \le j} U_{1l} + \sum_{l > j} U_{0l}, \qquad L_j(t) = \mathbb{E}(t - T_j)_+, \qquad V_i = \sum_{l < i} U_{1l} + \sum_{l > i} U_{0l}.$$

The mesh is increasing because $T_j - T_{j-1} \ge 1 - a_0 - a_1 > 0$. Each T_j has the conditional convolution given j successes. Set

$$h(x) = \mathbb{E}(x - U_0)_+ - \mathbb{E}(x - 1 - U_1)_+ - \mathbb{P}(1 + U_1 \le x), \qquad g_i = \mathbb{E}h(t - i + 1 - V_i).$$

Independence of the omitted coordinate gives $g_i = L_{i-1}(t) - L_i(t) - \mathbb{P}(T_i \le t)$. Here $L_0 = t$, $L_n = 0$ and $\mathbb{P}(T_0 \le t) = 1$. Multiplication by π_i and finite summation by parts yield

$$\begin{aligned} \mathbb{P}\left(\sum X_i \le H\right) &= (1-u)F_B(k) + uF_B(k+1) \\ &\quad + \sum_{j=1}^{n-1} (\pi_{j+1} - \pi_j) B_j - \sum_{i=1}^{n} \pi_i g_i, \quad B_j = L_j(t) - (t-j)_+ \ge 0. \end{aligned} \tag{58}$$

Indeed replacing L_j by $(t-j)_+$ in $\pi_0 + \pi_1 t + \sum(\pi_{j+1} - \pi_j)L_j$ telescopes to the stated interpolation. All CDF events are closed, including atoms.

Five kernels with four centered replacements.

Let $f(x) = \sin^2(\pi x)$ on $[0,1]$ and zero outside. It is continuously differentiable with derivative Lipschitz constant $2\pi^2$. We claim

$$h(x) \ge \eta f(x) \ge 0. \tag{59}$$

For $x < 1 - a_1$, $h(x) = \mathbb{E}(x - U_0)_+ \ge 0$, and Jensen gives $h(x) \ge x$ when $x \ge 0$. On $[0,1]$, $f(x) \le (5/2)x$: use $\pi^2 < 10$ up to $1/4$, the concave tangent $1/2 + \pi(x - 1/4)$ on $[1/4, 2/5]$, and $f \le 1$ thereafter. For $x \ge 1 - a_1 > a_0$, put $D_1 = 1 + U_1$ and $L_1 = \mathbb{E}(D_1 - x)_+$. Put–call parity gives $h(x) = \mathbb{P}(D_1 > x) - L_1 \ge 0$, since $D_1 \le 1 + A$ and $x > A$. When $1 - a_1 \le x \le 1$, $L_1 \ge 1 - x$ and

$$h(x) \ge \frac{x - A}{1 + A - x} L_1 \ge \frac{1 - A - d}{A + d}(1 - x) \ge \eta\pi^2(1 - x)^2 \ge \eta f(x).$$

The penultimate inequality uses $\eta\pi_+^2 d \le (1 - A - d)/(A + d)$. At $x = 0, 1$ both comparisons are weak, as required for atoms.

Freeze $V = V_{k+1}$ and let $V^{(j)}$ have the law of V_{k+1+j}. For $j > 0$ it replaces j copies of U_0 by U_1. With $f_j(x) = f(u - j - x)$, regroup the two positive changes as

$$\begin{aligned} &\mathbb{E}f_1(V^{(1)}) + \mathbb{E}f_2(V^{(2)}) - \mathbb{E}f_1(V) - \mathbb{E}f_2(V) \\ &\quad = \mathbb{E}(f_1 + f_2)(V^{(1)}) - \mathbb{E}(f_1 + f_2)(V) + \mathbb{E}f_2(V^{(2)}) - \mathbb{E}f_2(V^{(1)}). \end{aligned}$$

There is an identical grouping for the negative changes. Adjacent bumps have disjoint open supports and zero derivatives at the joins, so the derivative Lipschitz constant remains $2\pi^2$. For a centered Z, $|\mathbb{E}F(a + Z) - F(a)| \le \frac{1}{2}\operatorname{Lip}(F')\mathbb{E}Z^2$. Conditioning on all unchanged summands, and applying this twice to each replacement, bounds the total cost by $4\pi^2(s_0 + s_1)$. Hence

$$\sum_{i=k-1}^{k+3} g_i \ge \eta \left\{ \sum_{j=-2}^{2} \mathbb{E}f(u - j - V) - 4\pi^2(s_0 + s_1) \right\}.$$

These five indices are interior, because $|i - np| < \xi + 3 < 4$. No independence between different count sums is asserted.

The infinite translates of f sum to $\sin^2(\pi(u-V))$. An omitted nonzero translate requires $|V| > 2$, including endpoint collisions because the bump vanishes there. Thus the omitted part is at most $\mathbb{E}V^4/16$. With $\delta = k - np \in (-1-\xi, \xi)$, the frozen variance is

$$\tau = (nq - \delta - 1)s_0 + (np + \delta)s_1, \qquad (1 - 6/n)\lambda \le \tau \le (1 + 3/n)\lambda.$$

The lower estimate uses $c_- r \ge 1 + \xi$; the upper uses $r \ge 1/3$. Also $s_0 + s_1 \le \lambda/(nr)$ and $\mathbb{E}V^4 \le A^2\tau + 3\tau^2$. For either conditional noise Z, an independent copy satisfies $|Z - Z'| \le A + d < 2/3$. Concavity of sine and $\sin(2\pi/3) = \sqrt{3}/2 > 17/20$ give $\sin(\pi x) \ge (51/40)x$ on $[0, A+d]$; hence $1 - \cos(2\pi x) \ge 3x^2$ on $|x| \le A + d$. Therefore

$$1 - |\mathbb{E}e^{2\pi i Z}|^2 = \mathbb{E}\{1 - \cos(2\pi(Z - Z'))\} \ge 2\kappa \operatorname{Var} Z,$$

and $|\mathbb{E}e^{2\pi i Z}| \le e^{-\kappa \operatorname{Var} Z}$. Multiplying over the independent frozen summands shows $\mathbb{E}\sin^2(\pi(u - V)) \ge (1 - e^{-\kappa\tau})/2$. Altogether,

$$\sum_{i=k-1}^{k+3} g_i \ge \eta \left\{ \frac{1 - e^{-\kappa(1-6/n)\lambda}}{2} - \frac{A^2(1+3/n)\lambda + 3(1+3/n)^2\lambda^2}{16} - \frac{4\pi^2\lambda}{nr} \right\}. \tag{60}$$

The weights and the complete curvature tail.

For $k-1 \le i \le k+3$, the segment between i/n and p has variance product greater than ζ; indeed $\zeta < \min_{z = p_- - 4/N,\, p_+ + 4/N} z(1-z)$. Relative entropy is bounded by $(i - np)^2/(2n\zeta) < 8/(n\zeta)$, and the factorial remainder is greater than $-1/(12n\zeta)$. Stirling therefore gives

$$\sqrt{n}\pi_i > 2\phi_- \left(1 - \frac{8 + 1/12}{N\zeta}\right) > m_*.$$

Every other g_i is nonnegative. The reserve in (60) increases with n. Replace n by N first, then use $1 - e^{-x} \ge x - x^2/2 + x^3/6 - x^4/24$ for $x \ge 0$. Writing $\alpha = 1 - 6/N$, $b = 1 + 3/N$ and $z = \kappa\alpha$, we obtain

$$\sqrt{n}\sum_i \pi_i g_i \ge W(\lambda) := \eta m_* \left\{ \frac{z\lambda - z^2\lambda^2/2 + z^3\lambda^3/6 - z^4\lambda^4/24}{2} - \frac{A^2 b\lambda + 3b^2\lambda^2}{16} - \frac{4\pi_+^2\lambda}{Nr} \right\}. \tag{61}$$

Put $c = (np - q - t)_+$. For $a = t - j$ and $W_j = T_j - j$, centering gives

$$B_j = (\mathbb{E}|a - W_j| - |a|)/2 \ge 0, \qquad B_j \le \sqrt{T}/2, \qquad a_+ B_j \le T/4,$$

whenever $T \ge \mathbb{E}W_j^2$. The factorization $27x^4 - 256x + 256 = (3x-4)^2(3x^2 + 8x + 16)$ gives $B_j \le 27M/(256|a|^3)$ for $a \ne 0$, if $M \ge \mathbb{E}W_j^4$. The exact adjacent-mass formula is

$$\pi_{j+1} - \pi_j = \frac{np - q - j}{(n+1)v}\, \mathbb{P}\{\mathrm{Bin}(n+1, p) = j + 1\}.$$

Discard its nonpositive terms and bound the positive numerator by $(a+c)_+$. Since $c \le \bar{c} < 1$, at most one mesh point lies in each of $(-c, 0)$ and $[0, 1)$; together they cost $c\sqrt{T} + T/4$. The rest is bounded by the complete series $\sum i^{-2} < 5/3$, $\sum i^{-3} < 5/4$. Using (50) and $\sqrt{n}/(n+1)^{3/2} \le 1/n$ yields

$$\sqrt{n}\sum_j (\pi_{j+1} - \pi_j)B_j \le \frac{\beta_0}{nv^{3/2}} \left\{ \frac{T}{4} + c\sqrt{T} + \frac{27M}{256}\left(\frac{5}{3} + \frac{5c}{4}\right) \right\}. \tag{62}$$

A positive adjacent mass requires $j < np$, where its actual variance is at most $\lambda/q \le T = \lambda/q_*$ and $\mathbb{E}W_j^4 \le M = A^2T + 3T^2$. Moreover $J(\lambda) = j_1\lambda + j_0 \ge \sqrt{T}$, since $4j_0 j_1 \ge 1/q_*$.

Both signs of the normal correction.

Apply (49) at $k, k+1$ and interpolate. For $G(x) = \Phi((x-np)/\sqrt{nv})$, $G''(x) \le \phi(0)(np-x)_+/(nv)^{3/2}$. The interpolation remainder is

$$\sqrt{n}\{(1-u)G(k) + uG(k+1) - G(t)\} \le \frac{\phi(0)}{2nv^{3/2}} u(1-u)(u-w)_+.$$

If $H \le 0$, changing normal variance from v to $v+s$ is favorable. Using $u(1-u) \le 1/4$ and $u^2(1-u) \le 4/27$, the total normal cost is at most $\phi_+(\xi/8 + 2/27)/(nv^{3/2})$, and $c = (-q-w)_+ \le \bar{c}$. If $H \ge 0$, $c = 0$, interpolation costs at most $\phi_+(2/27)/(nv^{3/2})$, and the variance change costs at most $\phi(0)ws/(2v^{3/2})$ by $1-(1+x)^{-1/2} \le x/2$. The cubic correction is favorable by (51).

Define the nonnegative quadratics

$$S_c = \frac{T}{4} + cJ(\lambda) + \frac{27M}{256}\left(\frac{5}{3} + \frac{5c}{4}\right), \qquad E_- = \frac{\beta_0 S_{\bar{c}} + \phi_+(\xi/8 + 2/27)}{V_*},$$

$$E_+ = \frac{\beta_0 S_0 + \phi_+(2/27 + \xi\lambda/2)}{V_*}.$$

Because $V_*^2 < v_-^3$, combining (58)–(62) with these normal bounds makes the full excess strictly less than $-W(\lambda) + E_\pm(\lambda)/n$. It remains to compare these explicit polynomials.

Four rational endpoints finish the interval.

Set $P_\pm(\lambda) = W(\lambda) - E_\pm(\lambda)/N - \lambda/100$. Each $E_\pm$ is a convex quadratic, whereas direct differentiation gives

$$W''(\lambda) = -\eta m_* \left\{ \frac{z^2}{4}\left[(z\lambda - 1)^2 + 1\right] + \frac{3b^2}{8} \right\} < 0.$$

Thus $P_\pm$ is strictly concave. Substitution of the rational constants above gives the following exact rational lower bounds:

	$\lambda = 1/100$	$\lambda = 7/20$
P_-	.002314	.061304
P_+	.003341	.063680

All four are positive, proving $P_\pm > 0$ on the whole closed interval. Since $n \ge N$ and $E_\pm \ge 0$, this proves (48) for $1/100 \le \lambda \le 7/20$. The scalar checks used here involve only rational inequalities and $199/500 < \phi(0) < 2/5$, $25/8 < \pi < 22/7$; no parameter grid or tail truncation is involved.

The two positive-noise proofs have identical support and probability hypotheses, include their shared endpoint, and cover every phase. At $\lambda = 0$, conditional centering and $\mathbb{E}U^2 = 0$ give $U = 0$ almost surely, so (49) applies directly. This completes the proof of Theorem 6.1.

7 Composition and the finite boundary

Theorem 7.1 (Acquired-witness exclusion)**.** *Let $n \ge 700$ and let (P, H) attain the fixed-target maximum at C_{E} with $a(P, H) \ge 0$. Suppose $C_{n-1} \le C_{\mathrm{E}}$, the structural inequality* (22)*, and the complete entry Conditions 4.3 and 4.5 hold. Then such a pair cannot exist.*

Proof. Theorem 4.6 gives $nD \le 32.20665378$ and $|H| \le 1.83605680$ for this same oriented, compactly supported law. Section 5 gives its central gap, $p \in (3/8, 93/200)$, $a_0 + a_1 < 1/250$, amplitude $3/5$ and $\lambda < 7/20$. To check the normalization, put $B = \mathbf{1}_{\{X>0\}}$, $m_j = \mathbb{E}(X \mid B = j)$ and $\ell = m_1 - m_0 > 0$. Then

$$m_0 = -p\ell, \quad m_1 = q\ell, \quad 1 = pq\ell^2 + \theta, \qquad U = (X - m_B)/\ell,$$
$$s = \theta/\ell^2, \qquad pq + s = \ell^{-2}, \qquad (B + U - p)/\sqrt{pq + s} = X.$$

The marginal in Section 6 is exactly P, and its threshold is the original raw H, with $|H| < 46/25$. Positive noise gives $a(P, H) < -\lambda/100$; zero noise is Bernoulli and gives strictly negative excess by Schulz. Both contradict $a \ge 0$. □

Theorem 7.2 (Conditional sharp inequality)**.** *Assume a genuine finite base $C_j \le C_{\mathrm{E}}$ for $1 \le j \le 699$, the external structural and strict finite-binomial theorems used above, and the complete entry predicates at every least failure at $n \ge 700$. Then*

$$C_n < C_{\mathrm{E}} \ (n \ge 700), \qquad C_n \le C_{\mathrm{E}} \ (n \ge 1), \qquad \sup_{n \ge 1} C_n = C_{\mathrm{E}}.$$

Proof. If $C_n < C_{\mathrm{E}}$ fails above the base, choose the least failing index. Proposition 2.1 and the reflection argument in Section 2.2 supply an attained positive closed-event pair with $a \ge 0$. The finite base and minimality give the true predecessor $C_{n-1} \le C_{\mathrm{E}}$. The preceding theorem excludes the pair. The independent Bernoulli limit (2) gives equality for the global supremum. □

With a base only through 41, the same argument locates any first failure in $42, \dots, 699$. An unresolved earlier failure prevents the predecessor argument at later indices. Reflection also includes the exact atom rule:

$$\mathbb{P}(-S_n \le -H) - \Phi(-H/\sqrt{n}) = \Phi(H/\sqrt{n}) - \mathbb{P}(S_n < H) \ge \Phi(H/\sqrt{n}) - \mathbb{P}(S_n \le H).$$

The difference is the atom at H. The selected orientation acquires its own signed D budget; that budget is not copied under reflection.

Finite checks.

The entry predicates retain both quadratic passes, all four signed passes, 11264 nominal signed boxes and every incidence in the acceptor's internal 100-panel contract. The one-strip argument removes only downstream probability panels, noise tables and older local comparison branches. Rational scalar gates and actual-law controls accompany the continuous proofs; they do not replace them or the finite base. The v16 records report ten bounded checks passing, separately from full Fourier reproduction. The checking programs and interpreter remain part of the trusted numerical boundary; no full Lean formalization is claimed.

8 Discussion and reproducibility

The fixed target aligns the coefficient in extremality, contact and the predecessor estimate. Conservation prices departures from two points against one physical budget. After coarse entry, direct inequalities acquire one strip and two relative-noise estimates finish the comparison. Their bounds stay relative to λ down to zero; zero noise itself uses the strict binomial theorem.

The unresolved issue is finite: a genuine base through 699, or another argument closing the gap from a smaller base, is still needed. This work provides neither an independent four-summand proof nor a complete base through 41.

AI assistance.

AI systems were used extensively for mathematical exploration, exact checking programs, source and bibliography comparisons, and drafting and revision. Separate AI review roles examined component arguments; their agreement is not a substitute for mathematical proof or external human review. Failed arguments remain diagnostics, not premises. The authors are responsible for the content and presentation.

Sources and reproduction.

Sources and evidence are at `https://github.com/kolodziejek/berry-esseen-certificates`. This text integrates the selected v16 route from PR 5 at commit `1aa1dd015e3d73fbeecc1a7acbfa395cbb560b5f`. The original manuscript, numerical witnesses and reference baseline are preserved. A supplementary source map records each retained, replaced or omitted argument. Bounded reproduction uses `mathematics/simplification-v16/2026-10-29/run_checks.py`; isolated coarse-entry reproduction is a separate computation. Neither a document build nor the small checks establish the missing finite base.

A A uniform Fourier enclosure with explicit input hypotheses

We prove the compensated Fourier formula and the complete low- and high-frequency estimates used in Section 4. Every assertion concerns an actual marginal with the displayed moment and event bounds. Attainment and acquisition of those bounds are not premises of the Fourier proof itself; their use is ordered in that section.

For each integer $n \geq N \geq 42$, let $X = X_n$ be a real random variable with $\mathbb{E}X = 0$, $\mathbb{E}X^2 = 1$. Neither the law nor the real raw threshold $H = H_n$ must be fixed as n varies. Write

$$f(t) = \mathbb{E}e^{itX}, \quad \gamma = \mathbb{E}X^3, \quad e_n(t) = e^{-nt^2/2}(1 - in\gamma t^3/6), \quad V_n = f^n - e_n.$$

The two complete integrals are

$$I_0(H) = \int_0^{1/2} \frac{|\Im(e^{-itH}V_n(t))|}{t^3}\, dt, \qquad I_\infty = \int_{1/2}^{\infty} \frac{|V_n(t)|}{t^3}\, dt.$$

A.1 The inversion edge, including atoms and moving thresholds

Lemma A.1. *If a finite signed measure μ has mass and first moment zero and $\int x^2\, d|\mu|(x) < \infty$, then, for every $H \in \mathbb{R}$,*

$$\int (x - H)|x - H|\, d\mu(x) = -\frac{4}{\pi}\int_0^\infty \frac{\Im(e^{-itH}\widehat{\mu}(t))}{t^3}\, dt.$$

The integral on the right is absolutely convergent.

Proof. For real y, substitution and integration by parts give

$$\int_0^\infty \frac{ty - \sin(ty)}{t^3}\, dt = \frac{\pi}{4} y|y|.$$

The boundary terms vanish: near zero the numerator is $O(t^3)$, and at infinity it is $O(t)$. The integrand has the sign of y, since $v - \sin v \geq 0$ for $v \geq 0$. To calculate the constant, nonnegative Tonelli and elementary Laplace integration give

$$\int_0^\infty \frac{1 - \cos t}{t^2}\, dt = \int_0^\infty s\left(\frac{1}{s} - \frac{s}{s^2+1}\right) ds = \frac{\pi}{2}.$$

The absolute double integral of the compensated expression against $d|\mu|(x)$ equals

$$\frac{\pi}{4}\int (x-H)^2 d|\mu|(x) < \infty.$$

Absolute Fubini is therefore valid. The linear compensation disappears because $\int (x-H)d\mu = 0$, proving both assertions. No continuity of a CDF at H is involved. □

Apply this to the sum law minus the scale-$\sqrt{n}$ pushforward of the signed density

$$\phi(x)\{1+\gamma(x^3-3x)/(6\sqrt{n})\}.$$

This signed density has mass one, first moment zero and finite absolute second moment. Its transform after scaling is e_n. Gaussian integration by parts, using $q_z(x) = (x-z)|x-z|$ and $q_z''' = 4\delta_z$, yields

$$\varepsilon_n := \mathbb{E}q_{H/\sqrt{n}}(S_n/\sqrt{n}) - \mathbb{E}q_{H/\sqrt{n}}(Z) - \frac{2\gamma\phi(H/\sqrt{n})}{3\sqrt{n}} = -\frac{4}{\pi n}\int_0^\infty \frac{\Im(e^{-itH}V_n(t))}{t^3}\,dt.$$

Consequently the error $E_n = -n^{3/2}\varepsilon_n/2$ satisfies

$$|E_n| \le \frac{2\sqrt{n}}{\pi}\{I_0(H) + I_\infty\}. \tag{63}$$

This is an identity and bound for each n, H; it exchanges no limit in n with an integral or a varying law.

A.2 Sixth-order LOW with its actual coefficient bounds

Assume common bounds

$$\mu_4 := \mathbb{E}X^4 \le M \le 111/10, \quad |\gamma| \le g \le 37/20, \quad \mu_6 \le M_6,$$

and put

$$a = (3-\mu_4)/24, \quad z = (10\gamma - \mathbb{E}X^5)/120, \quad |a| \le A, \quad |z| \le Z.$$

Here A, Z, M_6 are nonnegative common ceilings, not quantities inferred from bounded third moments. Define

$$E = M_6/720 + g^2/72 + (1/2 + g^2/144)^3/5, \qquad B = A + Z/2 + E/4,$$

$$a_0 = \frac{N-1}{2N}, \quad b_0 = \frac{45-M}{96}(1-2/N) > 0, \quad \lambda_N = \sqrt{\pi/2}\left(\frac{N}{N-1}\right)^{3/2}.$$

Proposition A.2. *For all the laws and thresholds above,*

$$I_0(H) \le \frac{L_0 + L_H|H|}{\sqrt{n}} + \frac{L_2}{n},$$

where, for any $c_0 \ge \lambda_N$ and $c_1 \ge \lambda_N/2$,

$$\begin{aligned} L_0 &= c_0 Z + c_1 A g + 251 g^3/17280, \\ L_H &= c_0 A + 251 g^2/4800, \\ L_2 &= E/(2a_0^2) + B^2/(2b_0^3). \end{aligned}$$

Proof. Let $h(t) = \exp(-t^2/2 - i\gamma t^3/6)$. Taylor's integral remainder for e^{itX}, followed by expectation, gives its expansion through degree five with error at most $\mu_6 t^6/720$. For $w = -t^2/2 - i\gamma t^3/6$ and $0 \le t \le 1/2$,

$$|w| \le t^2(1/2 + g^2/144) < 1/6, \quad |e^w - 1 - w - w^2/2| \le |w|^3 e^{|w|}/6 \le |w|^3/5.$$

The first bound follows from $\sqrt{1/4 + y} \le 1/2 + y$; for the last use $e^{1/6} \le (1 - 1/6)^{-1} = 6/5$. Comparing the two polynomials gives

$$f - h = -at^4 - izt^5 + r_6(t), \qquad |r_6(t)| \le Et^6, \qquad |f - h| \le Bt^4. \tag{64}$$

The degree-six term in $w^2/2$ costs $g^2 t^6/72$; it has not been discarded.

Four integrations of $1 - \cos u \ge 0$ show $\cos u \le 1 - u^2/2 + u^4/24$ for every real u. Independence gives $\mathbb{E}(X - X')^4 = 2\mu_4 + 6$, and hence

$$|f(t)|^2 \le 1 - t^2 + (M + 3)t^4/12 \le 1 - (45 - M)t^2/48.$$

Thus $|f|, |h| \le e^{-(45-M)t^2/96}$ throughout the low interval. Every point on their complex joining segment obeys the same bound. Taylor's integral formula for the polynomial $v \mapsto v^n$ proves

$$f^n - h^n = n(f - h)h^{n-1} + R_2, \quad |R_2| \le \tfrac{1}{2}n(n-1)B^2 t^8 e^{-(n-2)(45-M)t^2/96}.$$

The leading remainder in (64) and this power remainder cost at most L_2/n after division by t^3 and integration: use $n - 1 \ge 2a_0 n$, $(n - 2)(45 - M)/96 \ge b_0 n$, $n(n-1) \le n^2$, and $\int_0^\infty t^3 e^{-At^2} dt = 1/(2A^2)$, $\int_0^\infty t^5 e^{-At^2} dt = A^{-3}$.

For the remaining leading terms let $b = n - 1$ and $v = b\gamma t^3/6$. Their imaginary part after rotation by Ht is bounded by

$$ne^{-bt^2/2}\{(Z + A|H|)t^5 + bAgt^7/6\}.$$

The Gaussian moments of orders two and four, after dividing by t^3, give $n\sqrt{\pi/2}b^{-3/2}(Z + A|H| + Ag/2)$. This is at most $\lambda_N(Z + A|H| + Ag/2)/\sqrt{n}$. Finally put $u = n\gamma t^3/6$. Direct sine and cosine remainders give

$$|\Im[e^{-itH}(h^n - e_n)]| \le e^{-nt^2/2}(|u|^3/6 + |H|tu^2/2).$$

The Gaussian moments of orders six and four respectively give $\sqrt{\pi/2}g^3/(432\sqrt{n})$ times 5, and $\sqrt{\pi/2}g^2|H|/(24\sqrt{n})$. Using $\sqrt{\pi/2} < 251/200$ gives precisely the two displayed rational coefficients. Every extended integrand was nonnegative. The only index comparisons were the displayed rational inequalities; in particular the law and H can vary arbitrarily with n. □

In the selected enclosure one has actual support $|X| \le R$ with $1 \le R \le 6$ and $\mathbb{E}|X|^5 \le E_5$. Then $M_6 = RE_5$ and $A = \max(2, M - 3)/24$ are valid. Its sharper Z is also licensed: for $Q = X^2 - \gamma X - 1$ direct multiplication gives

$$\mathbb{E}Q^2 = \mu_4 - 1 - \gamma^2, \quad \mathbb{E}X^5 = \gamma^3 + 2\gamma + \mathbb{E}[Q^2(X + 2\gamma)].$$

Reflect this auxiliary calculation if necessary so that $t = |\gamma| \ge 0$. For any $r \ge \max(R, 2g)$ the two signs of $\mathbb{E}X^5 - 10\gamma$ are at most

$$P_r(t) = r(M - 1) + (10 - 2M)t - rt^2 + t^3, \quad Q_r(t) = r(M - 1) + (2M - 10)t - rt^2 - t^3.$$

The coefficients of μ_4 before replacing it by M are $r-2t, r+2t \geq 0$. If $M \leq 5$, $P_r \geq Q_r$ and $P_r \leq r(M-1)+(10-2M)t-(r-g)t^2$, whose maximum on $[0,g]$ is at the clipped vertex used in the code. If $M > 5$, $P_r \leq r(M-1)$ and $Q_r \leq r(M-1)+(2M-10)t-rt^2$, again with the stated clipped vertex. For $r = 6$, $P_6'' = -12+6t < 0$ on $[0,g]$. If $M \leq 5$ and $P_6'(g) \geq 0$, its maximum is exactly $P_6(g)$; otherwise the preceding quadratic gives $6(M-1) + (10-2M)^2/[4(6-g)]$. For $M > 5$ the unrestrained vertex bound is $6(M-1)+(2M-10)^2/24$. Taking the minimum of these proved ceilings and $E_5 + 10g$, then dividing by 120, gives a valid coefficient ceiling Z. The radius is an actual support input, not a free tuning parameter.

A.3 All six HIGH pieces and their index gates

For independent copies of the same marginal suppose an actual event $\mathcal{E}$ satisfies

$$\mathbb{P}(\mathcal{E}) \geq \alpha > 0, \quad |X - X'| \geq d > 0 \text{ on } \mathcal{E}, \quad \mathbb{E}[(X-X')^2 \mid \mathcal{E}] \leq M_E.$$

Fix $b = 9/10$, $0 < r \leq 1/4$, and

$$h = 333/106 - (22/7)^3 r^2/6 > 0, \quad \kappa = \alpha(23/48)^2 d^2, \quad L(x) = \frac{x+2}{x^2+4x+2},$$

$$S_3 = \sum_{j=1}^{20} j^{-3} + 1/800, \quad S_5 = \sum_{j=1}^{20} (j-r)^{-5} + \frac{1}{4(20-r)^4}.$$

Let $c(u) = u/2 - (M+3)u^2/24$ and set

$$c_* = \min\{c(1/4), c(81/100)\}, \quad c_q = 1/8 - (M+3)/384, \quad s = [1-(M+3)53/600]/2.$$

Proposition A.3. *Suppose $c_* > 0$. For every $n \geq N$, $\sqrt{n} I_\infty$ is at most the sum of the following six terms:*

$$\begin{aligned}
T_1(n) &= 2\sqrt{n} e^{-nc_*}, \\
T_2(n) &= \frac{M_E}{4\pi^{3/2} h \sqrt{\alpha}} \{S_3 + 3S_5/(n\alpha h^2)\}, \\
T_3(n) &= \frac{M_E \sqrt{n}}{2\pi^2} e^{-n\alpha h^2 r^2}, \\
T_4(n) &= \frac{e^{-n\kappa b^2}}{2b^4 \kappa \sqrt{n}}, \\
T_5(n) &= \frac{\sqrt{n}}{2b^2} e^{-n\alpha(23/48)^2}, \\
T_6(n) &= \sqrt{n} e^{-n/8} \{2L(n/8) + \sqrt{M-1}/3\}.
\end{aligned}$$

If $s > 0$, T_1 may be replaced by its minimum with $2\sqrt{n} e^{-nc_q} L(ns/4)$. The fourth term may be replaced by its minimum with $\sqrt{n} e^{-n\kappa b^2} L(n\kappa b^2)/(2b^2)$. The common-floor sum at N is valid if

$$2Nc' > 1 \quad \text{for } c' \in \{c_*, \alpha h^2 r^2, \alpha(23/48)^2, 1/8\}.$$

For the optional first term also require $2Nc_q > 1$; for the optional fourth term require $N\kappa b^2 \geq 1/4$. If an optional gate fails, the simpler term remains available.

Proof. The triangle inequality separates $|f^n|$ and the entire comparator. On $[1/2, b]$, symmetrization gives $|f|^n \leq e^{-nc(t^2)}$. The function c is concave, so $c(t^2) \geq c_*$ and the integral of t^{-3} is less than two. This gives T_1. Also

$$c(u) - c_q - s(u-1/4) = (M+3)(u-1/4)(81/100-u)/24 \geq 0.$$

When $s > 0$ this chord may be integrated to infinity.

For completeness, the tail calculation used here is, for $x = nab^2 > 0$,

$$\sqrt{n}\int_b^\infty e^{-nat^2}t^{-3}dt \le \frac{\sqrt{n}e^{-x}}{2b^2}L(x). \tag{65}$$

Substitution and positive Fubini reduce the integral, without its prefactor, to $\int_0^\infty u(x+u)^{-1}e^{-u}du$. Put $P(u) = u^2 - 4u + 2$, $D = x^2 + 4x + 2 = P(-x)$. Exact polynomial division gives

$$\frac{1}{x+u} - \frac{-u^3 + (x+8)u^2 - (x^2+8x+20)u + x^3 + 8x^2 + 20x + 16}{D^2} = \frac{P(u)^2}{(x+u)D^2} \ge 0.$$

Integrate using the factorial moments of e^{-u}; the numerator's integral is $(x+3)D$. Subtracting the resulting lower bound times x from one gives $L(x)$. Applying (65) at endpoint $1/2$ and rate s proves the optional first term.

For $t \ge b$, symmetrization, the actual-event probability bound, and convex Jensen give

$$|f(t)|^n \le e^{-n\mathbb{E}\sin^2(t(X-X')/2)} \le \mathbb{E}[e^{-n\alpha\sin^2(tD/2)} \mid \mathcal{E}], \quad D = |X - X'|.$$

Positive Tonelli allows a separate partition for every $D > 0$. With $\ell = 2\pi/D$, use the zero cell $[0, \ell/2]$ and nonzero cells $\ell[j - 1/2, j + 1/2]$, $j \ge 1$. In a nonzero cell write $t = \ell(j+v)$. On its core $|v| \le r$, $|\sin \pi v| \ge h|v|$; on its wings $r \le |v| \le 1/2$ it is at least hr. These follow from $\sin x \ge x - x^3/6$ and monotonicity of sin on $[0, \pi/2]$, with directed bounds on π. For $|v| \le r$ Taylor's integral remainder gives

$$\{(j-v)^{-3} + (j+v)^{-3}\}/2 \le j^{-3} + 6(j-r)^{-5}v^2.$$

Extend the nonnegative Gaussian majorant over the whole real line. Its zeroth and second moments, the Jacobian $t^{-3}dt = D^2(4\pi^2)^{-1}(j+v)^{-3}dv$, and $\mathbb{E}[D^2 \mid \mathcal{E}] \le M_E$ give T_2. The two series tails are bounded by their decreasing integrals after 20. The total wing weight is at most $\int_{\ell/2}^\infty t^{-3}dt = D^2/(2\pi^2)$, giving T_3. Extending these sets below b only increases their nonnegative bounds.

In the zero cell put $u = tD/2 \le \pi/2$. Where $u \le 1/2$, $\sin u \ge 23u/24$, and $D \ge d$ yields rate κ. Extending the set to $[b, \infty)$ gives T_4 from $t^{-3} \le t/b^4$; (65) gives its optional improvement. Where $u \ge 1/2$, monotonicity gives $\sin u \ge 23/48$ and the total weight is at most $1/(2b^2)$, proving T_5. Endpoints have zero Lebesgue measure even if the conditional distance distribution has atoms.

Finally $|e_n(t)| \le e^{-nt^2/2}(1 + n|\gamma|t^3/6)$. The first part on the entire $[1/2, \infty)$ uses (65); the second uses $\int_{1/2}^\infty e^{-nt^2/2}dt \le 2e^{-n/8}/n$. Pearson's identity $\mu_4 - 1 - \gamma^2 = \mathbb{E}(X^2 - \gamma X - 1)^2 \ge 0$ gives T_6. No comparator tail is dropped.

For floor evaluation, $\sqrt{n}e^{-nc'}$ decreases for $2nc' > 1$, whereas n^{-1} and $n^{-1/2}e^{-nc'}$ decrease for every $c' > 0$. Also $L'(x) = -(x^2 + 4x + 6)/(x^2 + 4x + 2)^2 < 0$. Thus the optional first term decreases under its own gate, and the comparator decreases for $n \ge 5$. The sign of the derivative of $\sqrt{x}e^{-x}L(x)$ is the sign of

$$4 - 10x - 22x^2 - 13x^3 - 2x^4,$$

which is negative at $1/4$ and strictly decreasing on $(0, \infty)$. This proves the optional zero-cell gate. Every comparison is of a common scalar majorant, not of characteristic functions for different laws. □

A.4 The precise closed Fourier conclusion

Under the preceding moment, actual-event and scalar gates, let K_H be any complete HIGH sum evaluated at N, including its permitted optional improvements. Then

$$|E_n| \le \frac{2}{\pi}\{L_0 + K_H + L_2/\sqrt{N} + L_H|H|\} \le \frac{16}{25}\{L_0 + K_H + L_2/\sqrt{N} + L_H|H|\}. \tag{66}$$

The second inequality uses $\pi > 25/8$ and nonnegative coefficients. This is the constant Fourier budget and raw-threshold slope used in Section 4. Complete HIGH bounds for several valid windows may be minimized. The LOW coefficients must remain one complete valid triple.

For rational acceptance, positive denominator roots are rounded downward, numerator roots upward, and $\pi > 333/106$, $\sqrt{\pi} > 4431/2500$, $\pi < 22/7$ are directed correctly. These constants require no numerical transcendental oracle: polynomial division gives

$$22/7 - \pi = \int_0^1 \frac{x^4(1-x)^4}{1+x^2}\,dx > 0,$$

and the tangent addition formula with its first-quadrant branch gives

$$\pi = 16\arctan(1/5) - 4\arctan(1/239).$$

Four alternating terms for the first arctangent and the upper bound $\arctan(1/239) < 1/239$ give $\pi > 333/106 > 25/8$ by rational comparison. Squaring then verifies the two Gaussian constant enclosures used above. Its exponential helper replaces $x \ge 0$ by a downward rational $y \le \min(x, 64)$ and returns an upward rounding of $1/\sum_{j=0}^{64} y^j/j!$. This is at least e^{-x}. The floor argument applies to the underlying analytic bounds, not to a claim that this rounded helper itself preserves strict monotonicity.

The estimates are uniform over all laws and thresholds satisfying the stated inequalities at the actual index. They require no monotonicity of extremal Berry–Esseen constants, no fixed-law asymptotic limit, and no unproved disappearance of a lattice tail.

B Elementary Gaussian constants

Lemma B.1 (Elementary Gaussian constants)**.** *For the standard normal density,*

$$\|\phi''\|_\infty = \phi(0) < 2/5, \qquad \|\phi'''\|_\infty \le 2/\pi, \qquad 1/(4\pi) < 2/25.$$

Moreover $819/2000 < C_{\mathrm{E}} < 20487/50000 < 41/100$ *and* $\phi(0) > 0.39886 > 0.3988 = 997/2500$.

We give the exact comparisons used in Lemma B.1.

Proof. Since $\phi''(x) = (x^2-1)\phi(x)$, the norm on $x^2 \le 1$ is at most $\phi(0)$. On $x^2 \ge 1$, the maximum of $(x^2-1)e^{-x^2/2}$ is $2e^{-3/2} < 1$, attained when $x^2 = 3$. Thus the norm is exactly $\phi(0)$.

For completeness, the Gaussian transform can be obtained without an inversion theorem for the distribution being studied. Differentiating $J(t) = \int_{\mathbb{R}} e^{-x^2/2}e^{itx}\,dx$ under the integral and integrating by parts give $J'(t) = -tJ(t)$. The Gaussian integral, evaluated by squaring it and using polar coordinates, gives $J(0) = \sqrt{2\pi}$. Hence $J(t) = \sqrt{2\pi}e^{-t^2/2}$; applying the same formula with the variables interchanged gives

$$\phi(x) = \frac{1}{2\pi}\int_{\mathbb{R}} e^{-t^2/2}e^{-itx}\,dt.$$

Three differentiations are justified by the integrability of $|t|^3 e^{-t^2/2}$. Consequently

$$\|\phi'''\|_\infty \le \frac{1}{2\pi}\int_{\mathbb{R}} |t|^3 e^{-t^2/2}\,dt = \frac{2}{\pi}.$$

The last integral equals four after the substitution $w = t^2/2$ on the positive half-line.

Here are elementary rational comparisons for all numerical constants used in this section. Put $a = \arctan(1/5)$ and $b = \arctan(1/239)$. Tangent addition gives $\tan(2a) = 5/12$, $\tan(4a) = 120/119$ and $\tan(4a - b) = 1$. Also

$$0 < 4(1/5 - 1/(3\cdot 5^3)) - 1/239 < 4a - b < 4/5 < 1 < \pi/2.$$

The arctangent inequalities follow by integrating $1 - t^2 < 1/(1+t^2) < 1$, while $\pi/2 = 2\arctan(1) > 1$ follows by integrating $1/(1+t^2) > 1/2$ on $[0,1)$. It follows that $\pi = 16a - 4b$ and

$$\pi > 16(1/5 - 1/(3\cdot 5^3)) - 4/239 = 281476/89625 > 25/8.$$

This proves $\phi(0) < 2/5$ and $1/(4\pi) < 2/25$.

To bound C_{E} more closely, define

$$s_k(x) = \sum_{j=0}^{k-1} \frac{(-1)^j x^{2j+1}}{2j+1}, \quad \pi_- = 16s_6(1/5) - 4s_1(1/239), \quad \pi_+ = 16s_5(1/5) - 4s_2(1/239).$$

The remainder in the finite geometric expansion of $1/(1+t^2)$ has its stated alternating sign, so integration gives $\pi_- < \pi < \pi_+$. Let $r_- = 3162277660/10^9$ and $r_+ = 3162277661/10^9$. Squaring shows $r_-^2 < 10 < r_+^2$. The following comparisons are exact rational ones:

$$\left(\frac{819}{2000}\right)^2 < \frac{19 + 6r_-}{72\pi_+} < C_{\mathrm{E}}^2 = \frac{19 + 6\sqrt{10}}{72\pi} < \frac{19 + 6r_+}{72\pi_-} < \left(\frac{20487}{50000}\right)^2.$$

For example, each outer comparison follows by multiplying by its positive rational denominator; the displayed finite sums specify those denominators completely. All quantities are positive, so taking square roots proves the claimed bounds. In particular, $0.4095 < C_{\mathrm{E}} < 0.40974 < 0.41$. The same upper endpoint for π is smaller than $22/7$. Since $(0.39886)^2 < 7/44$, we also obtain $\phi(0) > 0.39886 > 0.3988$. □

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